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Received: 25 March 2025

Accepted: 27 November 2025

Published online: 05 December 2025

Cite this article as: Dejene F.M., Ayano Y.M., Feyisa D.W. *et al.* Knowledge distillation-based lightweight MobileNet model for diabetic retinopathy classification. *Sci Rep* (2025). <https://doi.org/10.1038/s41598-025-30893-7>

Fitsum Mesfin Dejene, Yehualashet Megersa Ayano, Degaga Wolde Feyisa, Taye Girma Debelee, Hiwot Taye Mekonnen, Girum Woldegebreal Gessesse, Zelalem Chimdesa Merga, Hasset Tamirat Molla & Destaw Mulie

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Knowledge Distillation-Based Lightweight MobileNet Model for Diabetic Retinopathy Classification

Fitsum Mesfin Dejene^{1*}, Yehualashet Megersa Ayano^{1*},
 Degaga Wolde Feyisa¹, Taye Girma Debelee^{1, 2*}, Hiwot Taye Mekonnen¹,
 Girum Woldegebreal Gessesse³, Zelalem Chimdesa Merga⁴,
 Hasset Tamirat Molla⁵, Destaw Mulie⁶

^{1*}Ethiopian Artificial Intelligence Institute, Addis Ababa, Ethiopia.

²Addis Ababa Science and Technology University, Addis Ababa, Ethiopia.

³St. Paul's Hospital Millennium Medical College, Addis Ababa, Ethiopia.

⁴Zewditu Memorial Hospital, Addis Ababa, Ethiopia.

⁵Addis Ababa University, Addis Ababa, Ethiopia.

⁶WAGA Ophthalmology Center, Addis Ababa, Ethiopia.

*Corresponding author(s). E-mail(s): fitsummesfin12@gmail.com;
yehualashet.megersa@aaii.et; tayegirma@gmail.com;

Contributing authors: degagawolde@gmail.com; Hiwottayemekonnen@gmail.com;
girum.wgebreale@sphmmmc.edu.et; zelalemchimdesa@gmail.com; hasset.tamirat@aau.edu.et;
destawmul@gmail.com;

Abstract

Diabetic retinopathy (DR) stands as a leading cause of global blindness. Early identification and prompt treatment are essential to prevent vision impairment caused by DR. Manual screening of retinal fundus images is challenging and time-consuming. Additionally, in low-income countries, there is a significant gap between the number of DR patients and ophthalmologists. Currently, machine learning (ML) and deep learning (DL) are becoming a viable alternative to traditional DR screening techniques. However, DL suffers a major limitation in resource-constrained devices because of its large model size and substantial computational demands. Knowledge distillation is a prominent technique for creating lightweight models, effectively transferring knowledge from a larger, complex model to a smaller, more efficient one without significant loss in performance. Therefore, in this research, a lightweight student model is proposed, which follows the MobileNet architectural design by utilizing depthwise separable convolutions. This design ensures efficient performance suitable for edge device deployment. For binary classification, our proposed model achieved an accuracy, precision, and recall of 98.38% on the APTOS 2019 dataset, whereas the proposed model achieved an accuracy of 93.03% for ternary classification on APTOS 2019.

Keywords: Diabetic Retinopathy, MobileNet, Knowledge Distillation, Lightweight Model, Classification

1 Introduction

Diabetic Retinopathy (DR), a microvascular complication of diabetes mellitus, has emerged as a leading cause of blindness among working-age adults (20-74 years) worldwide [1, 2]. According to the International Diabetes Federation (IDF), 451 million adults (18-99 years old) have diabetes worldwide, and by 2045, that number is expected to reach 693 million [3, 4]. Approximately half (49.7%) of people living with diabetes remain undiagnosed, significantly delaying critical ophthalmologist interventions [5]. Among diagnosed patients, global estimates indicate that 34.6% develop DR, with 10.2% progressing to vision-threatening stages [1].

The epidemiological situation in Ethiopia reflects concerning trends, with diabetes prevalence ranging from 2.0% to 6.5% across regions [6]. Recent studies report DR prevalence of 19.48% among Ethiopian diabetic patients, with 10.7% having vision-threatening DR (VTDR) [7, 8]. This poses substantial public health challenges given Ethiopia's severe ophthalmologist shortage (1:1,200,000 ratio) and limited eye care access (available to only 26% of the population) [9].

Current clinical guidelines recommend that type 1 diabetics undergo initial retinal examination 5 years post-diagnosis, while type 2 diabetics require immediate screening at diagnosis [10, 11]. However, poor compliance persists due to multifactorial barriers including limited health literacy, inadequate infrastructure, and insufficient insurance coverage [12, 13]. The diagnostic process itself remains labor-intensive, requiring ophthalmologists to manually screen fundus images for characteristic findings, including microaneurysms, intra-retinal hemorrhages, venous beading, exudates, and neovascularization [14]. With only 232,866 ophthalmologists globally serving millions of potential DR cases, this manual approach creates critical bottlenecks [15].

Lesions are symptoms of the severity level of DR [14]. Lesions can be broadly classified into four categories: soft and hard exudates (EX), hemorrhages (HM), and microaneurysms (MA). Due to the weakening of the vessel walls, MA, an early stage of DR, is identified by the appearance of tiny, round red spots on the retina. These dots have clear boundaries and are usually less than 125 micrometers in size [16]. On the other hand, unlike MA, HMs are recognized by the presence of large patches on the retina that have irregular edges and diameters greater than 125 micrometers. There are two forms of HM: flame and blot [17]. Flame refers to superficial areas, while blot indicates deeper ones. Hard EX are caused by blood leakage and appear as yellow patches on the retina. They have clear borders and span the outer layers of the retina [18]. Soft EX's are observable white ovals in the retina that result from swelling of nerve fibers [16]. Hemorrhages and MA frequently appear as red lesions, whereas both types of EX are usually seen in white lesions. The DR lesion details are shown in Figure 1. These lesions are critical for ML models, as they serve as the primary features from which the models learn.

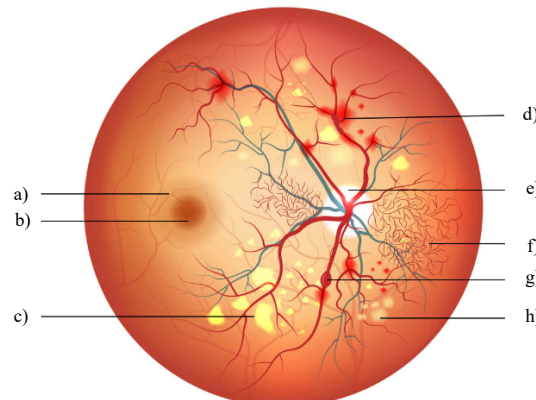


Fig. 1: Visualization of unhealthy retina with their respective lesion [19]: a) Macula, b) Fovea, c) Hard Exudate, d) Hemorrhages, e) Optic Disc, f) Abnormal Blood Vessels Growth, g) Aneurysms, h) Soft Exudate.

AI-enabled systems, particularly leveraging ML and DL, have become useful in automating DR screening by analyzing retinal fundus images for key lesions such as microaneurysms, hemorrhages, and exudates [20]. Among these, convolutional neural networks (CNNs) have demonstrated remarkable success, offering superior diagnostic accuracy, faster processing, and greater consistency compared to traditional manual screening methods, thereby reducing dependency on specialized ophthalmologists [21, 22]. These advancements facilitate large-scale, cost-effective screening programs, making them especially valuable in resource-limited settings like Ethiopia, where infrastructure and personnel constraints hinder traditional approaches [23].

However, despite their transformative potential, existing AI models face significant challenges, including large model sizes, high computational demands, and excessive memory footprints. These limitations hinder real-time usability, particularly on edge devices such as portable fundus cameras, mobile health platforms, or tele-ophthalmology setups, resulting in high energy consumption [20]. While tele-ophthalmology initiatives in Africa have provided partial solutions, delays persist due to the need for centralized professional interpretation of images [24]. Overcoming these barriers is critical for deploying AI-enabled DR screening in remote, low-resource environments.

This study addresses these challenges by developing a lightweight model for DR screening. By bridging the gap between AI advancements and practical deployment constraints, this research contributes to clinical practice in diabetes related vision care. To address these limitations, our research leverages knowledge distillation to develop a lightweight student model that achieves good diagnostic performance while significantly reducing computational requirements and model complexity. This makes the proposed solution well-suited for deployment in real-world, resource-constrained environments, thereby facilitating accessible and scalable DR screening.

The remaining sections of this paper are organized as follows. Section 2 provides an exploration of the related works. Section 3 discusses the methodology employed in our research. Section 4 presents the experimental setup, result analysis, and performance comparison. In Section 5, highlights the contributions of the proposed models. Finally, Section 6 summarizes the key findings and challenges and outlines future directions for further research.

2 Related Work

DR screening is an active research area focused on finding better techniques to assist physicians in diagnosing DR. As a result, several research papers have been published on DR screening, particularly in the context of binary and multi-class classification.

Anoop *et al.* [25] and Ishtiaq *et al.* [26] employed custom-designed CNN models for DR classification, which involve a large number of trainable parameters [25, 26]. Ishtiaq *et al.* [27] designed a custom CNN model to extract complex patterns of retinal lesions and used a classical ML classifier for classification. This combination of CNNs and ML classifiers improves overall performance by leveraging CNNs for feature extraction and ML classifiers for classification, despite the model's computational intensity. The CNN models proposed by Anoop *et al.* [25] and Ishtiaq *et al.* [26] are resource-intensive. In contrast, Bala *et al.* [28] developed a computationally efficient model resembling existing architectures [29], using four dense convolutional blocks and employing shortcut connections, which help maintain gradients during back-propagation. This model has 1.1 million parameters, making it lighter [28].

Pre-trained CNN models have also been used for binary classification of DR [30–33]. These models, including EfficientNet [32], ResNet [33], Inception-V3 [30], and DenseNet [31], are not architecturally identical but use the same fundamental CNN operations. While pre-trained models offer a substantial number of trainable parameters, training them with a relatively small number of samples raises concerns, particularly due to their high computational demands [30–33]. Beghriche *et al.* [34] compared the performance of pre-trained models on DR classification, finding that fine-tuned Xception outperformed DenseNet121 and MobileNetV2 [34].

The integration of custom-designed CNN models with techniques like active deep learning [35] and Siamese networks [36] offers a robust solution for image classification, especially when data is limited. Qureshi *et al.* [37] demonstrated active deep learning for DR classification, allowing the algorithm to select informative image patches, thereby optimizing model performance. Additionally, integrating

Siamese networks with custom CNNs, along with hierarchical clustering of image patches for feature extraction, further enhances performance [38]. This combined approach has proven effective in overcoming data limitations and improving classification outcomes [37].

Islam *et al.* [39] employed knowledge distillation to transfer knowledge from a teacher model, a fusion of ResNet152V2 and the Swin Transformer, to a student model, Xception, enhanced with a Convolutional Block Attention Module (CBAM). Despite using knowledge distillation, the teacher model remains resource-intensive, with 145.8 million parameters and 84.4 MB of memory, while the student model, although reduced, still retains 21.4 million parameters and 82 MB [39].

Some studies [40, 41] have addressed the complexities of DR classification by re-categorizing inseparable classes, which simplifies the model design and handling of data features, improving DR grading accuracy.

The VGG model, known for its hierarchical architecture, has been extensively applied to DR classification [42–44]. Khan *et al.* [44] enhanced VGG with stacked spatial pyramid pooling and network-in-network (NiN) layers, which improves scale invariance and non-linearity, important for identifying DR at varying image scales. However, VGG's computational demands can cause gradient vanishing during training, prompting the use of genetic algorithms (GA) as an optimization tool, though GA is also resource-intensive [45].

MobileNet and DenseNet offer computationally efficient alternatives for DR classification, especially in resource-constrained devices. MobileNet, designed for mobile and embedded applications, provides a lightweight option [46], while DenseNet's dense connectivity enhances supervision and reduces model complexity [29, 47]. According to Ayala *et al.* [48], DenseNet excels in parametric efficiency, while MobileNet's lighter structure makes it more suitable for mobile DR screening applications. InceptionV3, with its inception modules for multi-scale lesion detection, has demonstrated efficacy in DR grading by capturing features at varying scales. Although segmentation into smaller patches improves feature extraction, it can be suboptimal due to the convolutional operations' ability to capture localized information [49, 50]. Advanced models such as InceptionResNetV2 and graph neural networks (GNNs) further expand DR classification capabilities [51].

A novel semi-local centrality to identify influential nodes in complex networks by integrating multidimensional factors (SLCMF) [52]. Unlike traditional metrics, SLCMF integrates structural, social, and semantic factors, enhancing both accuracy and scalability. It employs distributed local subgraphs, redefines centrality using the average shortest path, and captures latent relationships through semantic graph embedding. On the other hand, augmentation of the binary grey wolf optimization through quantum computing methodology was used for vision-threatening DR [53].

Study [54] identified Lipocalin-2 (LCN2) as a key mediator of neuroinflammation in retinal ischemia-reperfusion injury, suggesting its potential as a biomarker for glaucoma. Additionally, study [55] linked endocrine disruptors to diabetes through mitochondrial dysfunction, highlighting disruptions in oxidative phosphorylation and ROS generation. Technological advances include a CRDS-based breath analyzer [56] for non-invasive metabolic monitoring and deep learning methods like CS-Net [57] and AM-Net [58] for real-time ultrasound super-resolution imaging. In metabolic regulation, [59] emphasized the role of selenoproteins, while [60] characterized pembrolizumab-induced uveitis as a treatable immune-related adverse event. Generally, these works [54–60] underscore interdisciplinary progress in pathophysiology and precision diagnostics.

Furthermore, ML techniques offer promising accuracy for automated glaucoma detection by analyzing retinal images through preprocessing, feature extraction, and classification, providing valuable clinical support in identifying glaucomatous symptoms [61]. A hybrid model, ML and Nature-Inspired Model for Coronavirus (MLNI-COVID-19), combines ML and nature-inspired algorithms to enhance the classification and optimization of brain Magnetic Resonance Imaging (MRI) scans in COVID-19 patients, demonstrating improved diagnostic accuracy, sensitivity, and specificity [62]. However, computational complexity remains challenging in a resource-constrained environment. Soft-computing-based gravitational search optimization is used for feature selection, to eliminate unnecessary features, enhance performance, and reduce computational complexity for glaucoma predictions [63]. On the other hand, a genetic algorithm-based differential evolution-based multi-objective feature selection approach is used to extract only important features [64].

3 Methodology

3.1 Dataset

DL algorithms rely heavily on large datasets to understand image patterns that depict infections or lesions, as well as normal conditions. Training ML or DL models with substantial and high-quality datasets enhances model performance.

The Asian Pacific Tele-Ophthalmology Society (APTOS 2019) [65] is one of the most widely used publicly available retinal fundus image datasets, published on Kaggle. It contains 3662 retinal fundus images with varying resolutions, collected from Aravind Eye Hospital in India. The dataset was specifically designed for DR screening, making it highly suitable for training and evaluating DR detection models. The class distribution of the dataset is shown in Figure 2, providing insights into the prevalence of different DR severity levels within the dataset.

In addition to the APTOS 2019, the primary dataset was collected from local eye clinic centers, including WAGA Ophthalmology Center, Biruh Vision Specialized Eye Care Center, and KENESER Specialized Eye Clinic. 3D OCT-1 Maestro 2 and DRI OCT Triton of the Topcon funduscopy machine are used to capture retinal fundus from the eye care center. The collected images first underwent an expert-based filtering process to remove low-quality or noisy data. In the second stage, the remaining images were independently annotated by two experienced ophthalmologists. The dataset is categorized into three classes: No DR (normal retinal fundus), NPDR (non-proliferative diabetic retinopathy), and PDR (proliferative diabetic retinopathy). The class distribution is presented in Figure 2.

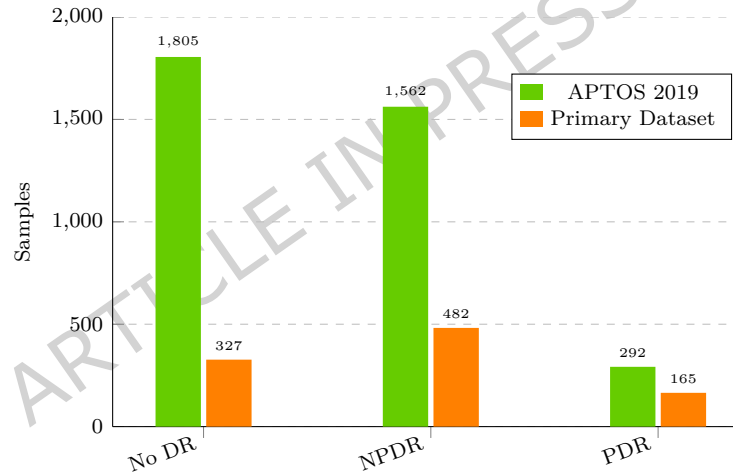


Fig. 2: APTOS 2019 and Primary datasets class distribution

The samples of retinal fundus images for each class are shown in Figure 3, highlighting the lesions or features that distinguish one class from another.

3.2 Data Preprocessing

The image pre-processing steps involve noise removal, quality enhancement, and preparation of retinal fundus images to be suitable for the model.

Retinal fundus images often contain black borders around the actual retina, which do not contribute useful features for class differentiation. To address this, the cropping process converts the image to grayscale, applies a threshold (set to 7) to create a binary mask, and retains only the areas with pixel values above the threshold, effectively removing irrelevant dark regions [66]. Additionally, a bi-linear interpolation down-scaling algorithm is used to reduce computational load and memory requirements, while adjusting the image size to fit the model's input size. Since fundus images are typically in RGB format, all channels are retained to capture comprehensive features, although the green channel is often

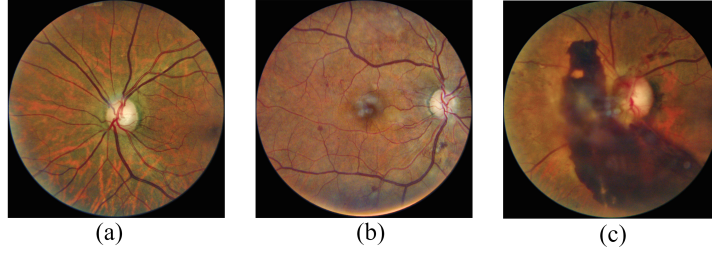


Fig. 3: Severity level of diabetic retinopathy [65]: (a) No DR; (b) NPDR; (c) PDR

most useful for highlighting blood vessels. Furthermore, contrast-limited adaptive histogram equalization (CLAHE) with a clipLimit of 2.0 and tileGridSize of (8, 8) is applied channel-wise to enhance the contrast of the image. CLAHE improves primary contrast, making features in the retinal image more visible and easier for the model to analyze. The pre-processing techniques are illustrated in Figure 4 and Algorithm 1

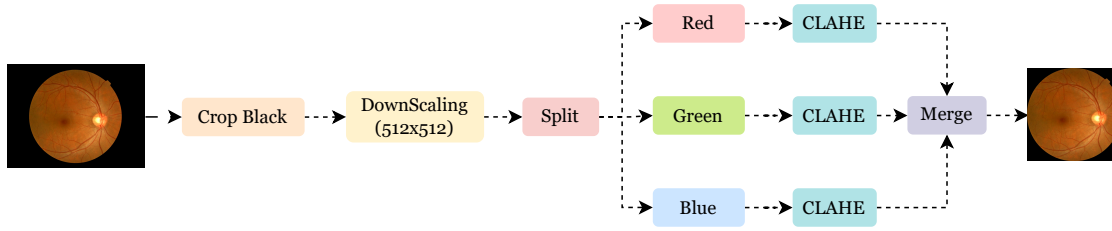


Fig. 4: Retinal data pre-processing

Algorithm 1 Pseudocode for data processing and model training

- 1: **procedure** DATA PRE-PROCESSING
 - 2: Cropping black pixel located at the edge.
 - 3: Down-scale the retinal fundus image from various sizes to 512x512.
 - 4: Split the retinal image in three channels (Red, Green, and Blue).
 - 5: Apply CLAHE on each channel.
 - 6: Merge each channel.
 - 7: **end procedure**
-

The dataset was initially split into two subsets: 85% of the pre-processed data as the training dataset and 15% as the testing dataset. The training dataset was then further divided into 85% as the training set and 15% as the validation set. Following the data splitting steps, data augmentation was applied on the training set to enhance dataset diversity, thereby improving model generalization and performance. This process was consistently applied to the primary dataset and the APTOS 2019 datasets for model training.

3.3 Model

MobileNet is a family of lightweight CNN architectures optimized for efficient deployment on mobile and embedded devices [46]. In this study, a MobileNet-like structure is chosen due to its ability to significantly reduce computational complexity and memory footprint through the use of depthwise separable convolutions. This design makes the model highly suitable for resource-constrained devices such as smartphones or portable fundus cameras. Moreover, MobileNet has demonstrated competitive performance

in various computer vision tasks, including medical imaging, while maintaining an architecture. These characteristics align with the goal of developing a lightweight, reliable DR screening model deployable in remote or low-resource settings. To further reduce the model's computational complexity, knowledge distillation (KD) is used to transfer knowledge from a larger, pre-trained "*Teacher*" model to a smaller "*Student*" model. This process ensures that the smaller model achieves comparable performance while significantly reducing computational requirements, making it ideal for resource-constrained devices [67] [68]. Figure 5 shows the architecture of the "*Teacher*" model and "*Student*" model.

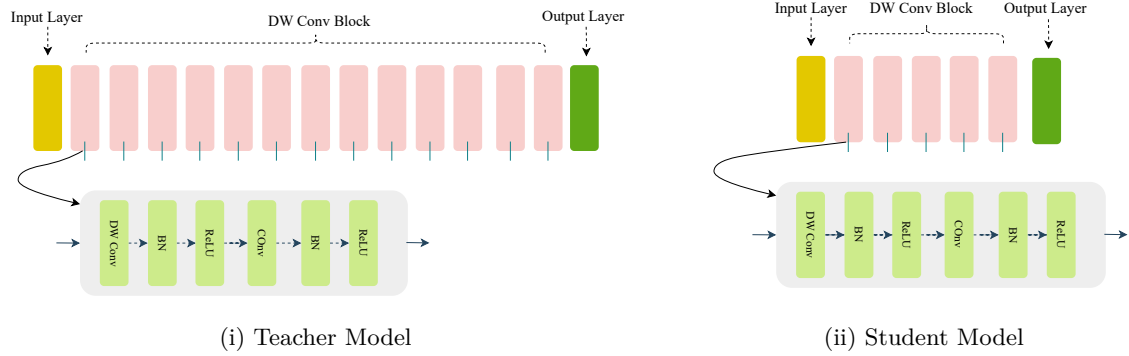


Fig. 5: *Teacher* and *Student* model architectural differences

A lightweight *student* model was developed from MobileNet, by the principle of simplifying the teacher network by reducing the number of network layers and the sizes of filters, as outlined by Gou *et al.*[67]. This alignment also minimizes the "model capacity gap", where a significant difference can hinder the *Student* model's ability to effectively gain knowledge from the *Teacher* [67]. Then deeper blocks were systematically trimmed, filter sizes adjusted, stride patterns modified, and a custom fully connected classification head was used to achieve an optimal balance between computational complexity and predictive performance. These choices were specifically tailored for high-resolution (512×512) medical images and resource-constrained deployment scenarios. Reducing the deeper layers in a *Student* model can help maintain important feature extraction ability while balancing computational efficiency and performance. The deeper layer frequently contains complicated, high-level information; yet, these layers can be computationally intensive. Therefore, by removing deep layers and building the shallower layers, which capture the core structural and low-level elements, the *Student* model achieves a good balance between maintaining useful information and minimizing the model's overall size and latency. Our approach aligns with Wang *et al.* [69], who explained that simplified models focus on retaining the main features while offloading complex, task-specific details to improve interpretability and performance on resource-constrained devices. The detailed *Student* model architecture is shown in Figure 6 and Table 1.

The distillation of knowledge on MobileNet involves key components and steps as mentioned in Algorithm 2. The distillation process employs a composite loss function combining the standard task-specific loss with a distillation loss that encourages the Student to mimic the Teacher's output distribution. The task-specific loss is computed using Categorical Cross-Entropy, while the distillation loss is calculated using the Kullback-Leibler (KL) divergence applied to softened probability distributions from both models.

The *Teacher* model is a robust, pre-trained neural network that serves as the source of knowledge. It generates "soft labels" which are *logits*, probability distributions over classes, providing richer and more information than hard labels. On the other hand, the *Student* model is a smaller, lightweight network designed to replicate the performance of the *Teacher* model. The distillation loss is based on a composite loss function that integrates both soft and hard losses. Figure 7 shows the distillation process and how losses are calculated.

The predictions of the *Teacher* and *Student* models are softened using a temperature parameter T , as described in Eqs. (1) and (2). The application of the *SoftMax* function with T smooths the probability

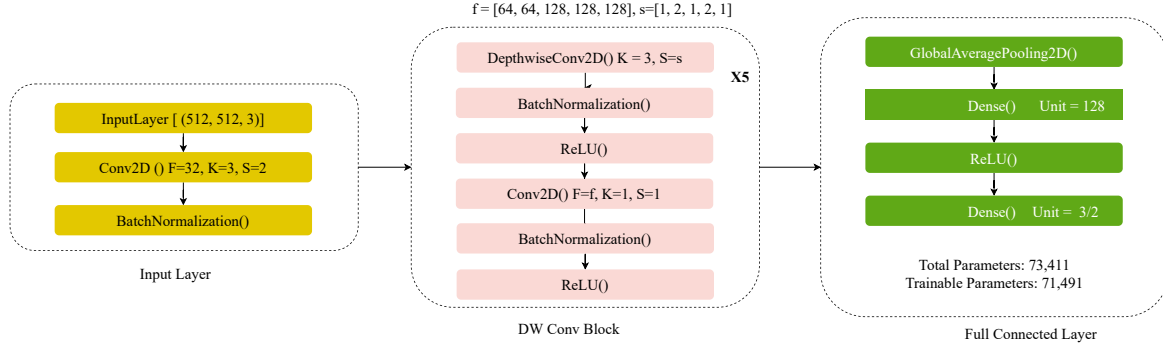


Fig. 6: Architecture of the proposed *Student* model

Algorithm 2 Pseudocode for Knowledge Distillation

- 1: **procedure** TEACHER MODEL TRAINING
 - 2: Build MobileNet pre-trained model.
 - 3: Train the model on the retinal fundus dataset.
 - 4: Evaluate the performance of the model.
 - 5: **end procedure**
 - 6: **procedure** DISTILLATION PROCESS
 - 7: Build *Student* model.
 - 8: Define Distiller with the correct configuration.
 - 9: Train the student model through distiller for knowledge transfer.
 - 10: Evaluate the performance of the *Student* model.
 - 11: **end procedure**
-

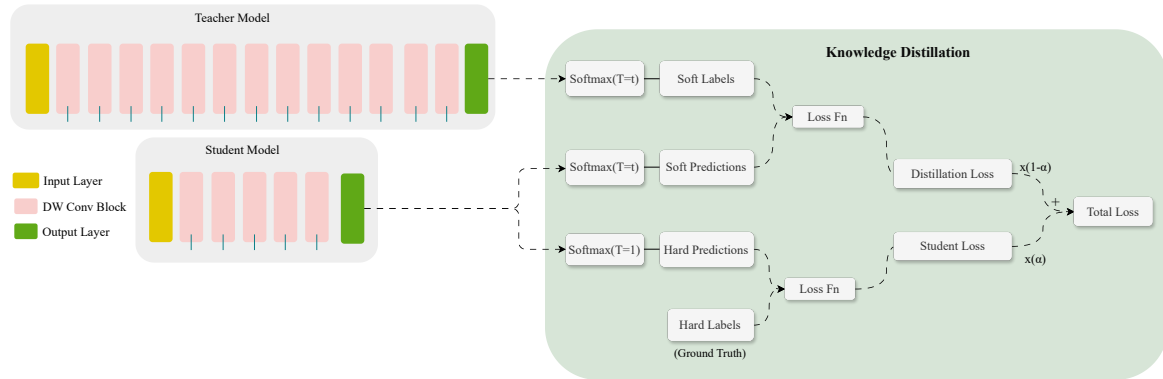


Fig. 7: MobileNet knowledge distillation process

distributions of the *Teacher*, making them less sharp (peaky) and more informative. This softened output allows the *Student* model to better capture important patterns during training.

$$\text{softLabels} = \text{softMax} \left(\frac{\text{Logits}_{\text{teacher}}}{T} \right) \quad (1)$$

$$\text{softPredictions} = \text{softMax} \left(\frac{\text{Logits}_{\text{student}}}{T} \right) \quad (2)$$

To quantify how well the *Student's* prediction distribution approximates the *Teacher's* softened prediction distribution, the Kullback-Leibler (KL) Divergence is used. Therefore, the distillation loss is computed by applying KL Divergence to the softened labels (Eq. 1) and softened predictions (Eq. 2):

$$\text{distillation_loss} = \sum_x \text{SoftLabels}(x) \log \left(\frac{\text{SoftPredictions}(x)}{\text{SoftLabels}(x)} \right) \quad (3)$$

Here, x represents the output logit for each class. The KL divergence quantifies how closely the Student’s softened probability distribution approximates that of the Teacher.

The total loss for training the *Student* model combines the task-specific loss (`student_loss`), which uses `CategoricalCrossentropy`, and the distillation loss. The total loss is defined in Eq. (4):

$$\text{loss} = \alpha \cdot \text{student_loss} + (1 - \alpha) \cdot \text{distillation_loss} \quad (4)$$

Here, α is a hyperparameter that controls the trade-off between learning from the ground truth labels (`student_loss`) and learning from the *Teacher*’s knowledge (`distillation_loss`). When $\alpha \rightarrow 1$, the *Student* model relies more heavily on the task-specific loss; when $\alpha \rightarrow 0$, the *Student* model relies predominantly on the *Teacher*’s softened knowledge.

While the `Distiller` class allows flexibility in selecting α and temperature T , systematic hyperparameter tuning was conducted using Keras-Tuner’s Hyperband algorithm to identify optimal values. The search space for α was set to $[0.1, 0.9]$ with increments of 0.1, and for temperature T to $[0, 50]$ with increments of 5, optimizing for validation accuracy. The best-performing configuration was found to be $\alpha = 0.5$ and $T = 10$, and used for the knowledge distillation from the *Teacher* model, leading to improved *Student* model performance.

Table 1: Detailed architecture of the proposed student model.

Layer Type	Output Shape	Parameters
Input Layer	(512, 512, 3)	0
Conv2D (3×3, 32 filters, stride 2)	(256, 256, 32)	896
Batch Normalization	(256, 256, 32)	128
ReLU Activation	(256, 256, 32)	0
Block 1 (64 filters, stride 1)		
Depthwise Conv2D (3×3)	(256, 256, 32)	320
Batch Normalization	(256, 256, 32)	128
ReLU Activation	(256, 256, 32)	0
Pointwise Conv2D (1×1, 64 filters)	(256, 256, 64)	2,112
Batch Normalization	(256, 256, 64)	256
ReLU Activation	(256, 256, 64)	0
Block 2 (64 filters, stride 2)		
Depthwise Conv2D (3×3)	(128, 128, 64)	640
Batch Normalization	(128, 128, 64)	256
ReLU Activation	(128, 128, 64)	0
Pointwise Conv2D (1×1, 64 filters)	(128, 128, 64)	4,160
Batch Normalization	(128, 128, 64)	256
ReLU Activation	(128, 128, 64)	0
Block 3 (128 filters, stride 1)		
Depthwise Conv2D (3×3)	(128, 128, 64)	640
Batch Normalization	(128, 128, 64)	256
ReLU Activation	(128, 128, 64)	0
Pointwise Conv2D (1×1, 128 filters)	(128, 128, 128)	8,320
Batch Normalization	(128, 128, 128)	512
ReLU Activation	(128, 128, 128)	0
Block 4 (128 filters, stride 2)		
Depthwise Conv2D (3×3)	(64, 64, 128)	1,280
Batch Normalization	(64, 64, 128)	512
ReLU Activation	(64, 64, 128)	0
Pointwise Conv2D (1×1, 128 filters)	(64, 64, 128)	16,512
Batch Normalization	(64, 64, 128)	512
ReLU Activation	(64, 64, 128)	0
Block 5 (128 filters, stride 1)		
Depthwise Conv2D (3×3)	(64, 64, 128)	1,280
Batch Normalization	(64, 64, 128)	512
ReLU Activation	(64, 64, 128)	0
Pointwise Conv2D (1×1, 128 filters)	(64, 64, 128)	16,512
Batch Normalization	(64, 64, 128)	512
ReLU Activation	(64, 64, 128)	0
Fully Connected Layers		
Global Average Pooling 2D	(128,)	0
Dense (128 units, ReLU)	(128,)	16,512
Dense (2 units)	(2,)	258
Dense (3 units)	(3,)	387
Parameter Summary		
Dense (2 units)	Trainable	71,362
	Non-trainable	1,920
	Total	73,282
Dense (3 units)	Trainable	71,491
	Non-trainable	1,920
	Total	73,411

4 Experiment and Result Analysis

4.1 Experimental Setup

The proposed model was trained and tested on a machine equipped with hardware specifications of 13th Gen Intel(R) Core(TM) i7-13700H, 2400 MHz, and 8GB NVIDIA-SMI 546.12 GPU, CUDA 12.3. The proposed model was implemented using the Keras API on the TensorFlow framework version 2.10.0.

4.2 Performance Evaluation Metrics

Performance metrics are quantitative measures used to evaluate our model's effectiveness on the dataset. The metrics formula used in our study is shown as follows;

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN} \quad (5)$$

$$Precision = \frac{TP}{TP + FP} \quad (6)$$

$$Recall = \frac{TP}{TP + FN} \quad (7)$$

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall} \quad (8)$$

$$AUC = \int_a^b F(x) d(x) \quad (9)$$

4.3 Result Analysis

4.3.1 Teacher Model Selection

This research utilizes MobileNet, a family of lightweight CNNs optimized for efficient use on mobile and embedded devices. Both MobileNet and MobileNetV2 are evaluated to find the best *Teacher's* model that can effectively lead to the development of a lightweight, accurate *Student* model. During the *Teacher* model selection phase, the models are evaluated on ternary classification: No-DR, Non-proliferative DR, and Proliferative DR. The detail of model configuration is shown in Table 2 and various values of α , which determine the width multiplier. Our experiments showed that MobileNet with α value of 0.25 achieved the optimal combination of performance and efficiency, making it the ideal *Teacher* model for our research. Table 4 and Figure 8 show the results of MobileNet with MobileNetV2 at various α values. These visualizations provide a clear overview of each model's strengths and help us to identify a high-performing *Teacher* model, which will later be refined into an even lighter *Student* model suitable for real-time application in a computational constraint device.

Table 2: Summary model configurations for *Teacher*, *Student* model, and *Distiller*

Parameter	Values
Input Shape	512x512
Batch Size	8
Initial Learning Rate	0.0001
Optimizer	Adam
Alpha α for Distiller	0.5
Temperature T for Distiller	10

Furthermore, the model training setup and configuration were thoughtfully designed to ensure smooth convergence and optimal generalization of the proposed student model enhanced through knowledge

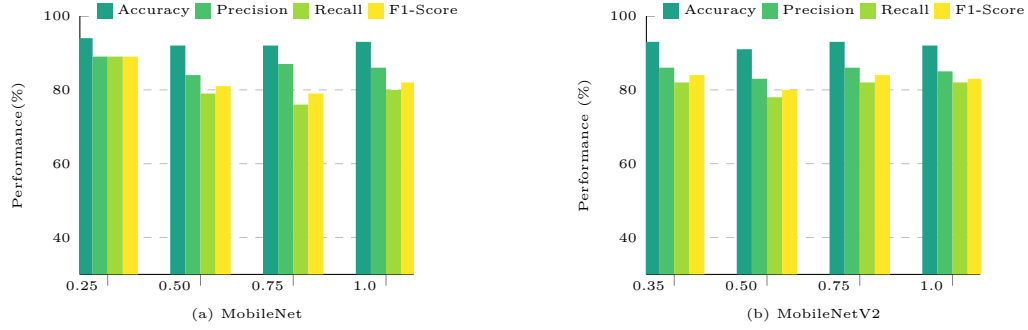


Fig. 8: The performance of MobileNet and MobileNetV2 at different α values on APTOS 2019 dataset

distillation. To support effective learning and systematically monitor the model's performance, several callback mechanisms were integrated during training. Specifically, a ReduceLROnPlateau callback was employed to dynamically adjust the learning rate by a factor of 0.2 when the validation loss failed to improve for three consecutive epochs, with the minimum learning rate capped at 1×10^{-12} . To mitigate overfitting, EarlyStopping was applied with a patience of 13 epochs, ensuring that the model reverted to the best-performing weights observed during training. The complete details of these configurations are summarized in Table 3.

Table 3: Model Training Configuration and Callback Settings

Parameter/Callback	Value/Description
ReduceLROnPlateau	Factor = 0.2, Patience = 3, Minimum LR = 1×10^{-12}
EarlyStopping	Patience = 13 epochs, Restore Best Weights = True

Table 4: Comparison of *Teacher* models selection on ternary classification on APTOS 2019 dataset

Model	Alpha (α)	Trainable Parameter	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)
MobileNet	0.25	279,635	94	89	88	89
	0.50	950,691	92	84	79	81
	0.75	2,014,195	92	87	76	79
	1	3,470,147	93	86	80	82
MobileNetV2	0.35	724835	93	86	82	84
	0.50	1,016,387	91	83	78	80
	0.75	1,684,131	93	86	82	84
	1	2,552,579	92	85	82	83

4.3.2 Binary Classification

In binary classification, our objective is to evaluate the performance of the model in classifying retinal fundus images into two groups: 'No_DR' and 'DR'. The model performed well in the binary classification task, indicating that it is capable of accurately distinguishing the two classes. This high degree of separability demonstrates the model's ability to efficiently differentiate normal retinal fundus images

from those containing DR features. The *Teacher* model demonstrates superior performance on APTOS 2019 and primary datasets compared to the *Student* models, including both the *Student* with KD and without KD. Notably, the *Student* model with KD achieves comparable performance to the *Teacher* model across these datasets. This outcome shows the effectiveness of KD techniques in successfully transferring insights from the *Teacher* model to the *Student* model. Additionally, training the model on the primary dataset begins with initializing the model with the weights gained from the APTOS 2019 dataset. This approach adopts the model's generalization capacity from the APTOS 2019 dataset to a primary dataset, assuring its robustness and effectiveness in deployment for a local eye clinic.

Table 5 presents the performance of the *Teacher* and *Student* models, with and without knowledge distillation, on the APTOS 2019 dataset. Similarly, Table 6 shows primary datasets. The *Student* model trained with KD consistently outperformed students without KD, achieving an accuracy of 98.36% on APTOS 2019 and 93.20% on the primary dataset. These results demonstrate the effectiveness of KD in enhancing model generalization and accuracy, especially under limited data conditions. While all models performed best on APTOS 2019, the performance drop on the primary dataset is likely due to the small dataset and variability. Nonetheless, KD proved beneficial in transferring knowledge from the *Teacher* to the *Student* model. Additionally, the model's performance is shown in detail in Figure 9.

Table 5: Models Performance of Binary Classification on APTOS 2019 Dataset

Model	Averaging	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)
Teacher	macro avg	99.45	99.45	99.45	99.45
	weighted avg	99.45	99.45	99.45	99.45
Student without KD	macro avg	94.73	94.73	94.73	94.73
	weighted avg	94.73	94.733	94.73	94.73
Student with KD (Proposed Model)	macro avg	98.36	98.36	98.36	98.36
	weighted avg	98.36	98.36	98.36	98.36

Table 6: Model's performance of binary classification on a primary dataset

Model	Averaging	Accuracy (%)	Precision (%)	Recall(%)	F1 Score(%)
Teacher	macro avg	93.88	92.65	93.88	93.21
	weighted avg	93.88	94.05	93.88	93.92
Student without KD	macro avg	85.71	83.79	86.73	84.69
	weighted avg	85.71	87.28	85.71	86.01
Student with KD (Proposed Model)	macro avg	93.20	93.24	91.33	92.18
	weighted avg	93.20	93.20	93.20	93.12

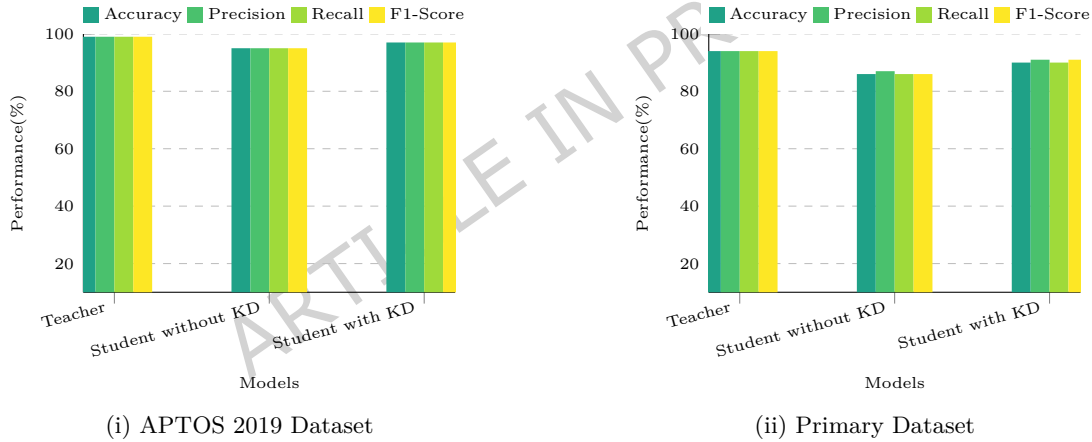
Cross-validation is a resampling technique used to assess the performance of a machine learning model by partitioning the dataset into multiple subsets. This method ensures that the model is evaluated on unseen data, reducing the risk of overfitting and providing a more reliable estimate of its generalization performance.

Furthermore, cross-validation is employed to evaluate the robustness of the proposed model as shown in Table 7. Given the high class imbalance in the dataset, the StratifiedKFold method is utilized to maintain the original class distribution across training and validation splits, ensuring a fair evaluation.

The model's learning curve gives useful information about the pattern of the *Teacher* model and *Student* model with KD and without KD in the training stage, indicating if the model has reached the optimal performance. It also helps in detecting signs of over-fitting or under-fitting. Additionally, it

Table 7: Comparison of *Student* with knowledge distillation model cross-validation on APTOS 2019 and Primary Dataset

Dataset	Fold (k)	Accuracy(%)	Precision(%)	Recall (%)	F1 Score (%)
APTOS 2019	1	98.36	98.36	98.36	98.36
	2	97.45	97.46	97.45	97.45
	3	97.64	97.64	97.64	97.64
	4	97.09	97.09	97.09	97.09
	5	97.82	97.83	97.82	97.82
	Mean	97.67	97.68	97.67	97.67
	Std	± 0.46	± 0.47	± 0.46	± 0.46
Primary Dataset	1	89.12	89.12	89.12	89.12
	2	93.20	93.20	93.20	93.12
	3	88.44	88.81	88.44	88.01
	4	92.52	92.91	92.52	92.60
	5	91.84	91.81	91.84	91.74
	Mean	91.04	91.17	91.04	90.92
	Std	± 1.89	± 1.87	± 1.89	± 2.01

**Fig. 9:** Comparison of models' performances on APTOS 2019 and primary datasets for binary classification

indicates the model's learning pattern, indicating whether it is failing to learn or progressively improving over time. Therefore, the learning curve of our models demonstrates strong performance and convergence to an optimal performance. Additionally, the curve does not show any signs of under-fitting or over-fitting, indicating a well-fitted model. However, the learning curve of the models on the primary dataset shows slight fluctuations, indicating difficulty in understanding the unique aspects of the primary data, particularly for the *Student* model without KD. Despite these fluctuations, the curve gradually converges, indicating that the model has been successfully trained to identify DR from normal retinal fundus images. Figures 10, 11 shows the learning curve of student model with KD, where as Figure 18, 19 shows the learning curves of the *Teacher* and *Student* models without knowledge distillation, on the APTOS 2019 and primary datasets, respectively.

A confusion matrix provides a clearer picture of class-wise model performance, especially on imbalanced datasets, making it essential for targeted model improvements. Therefore, our model's confusion

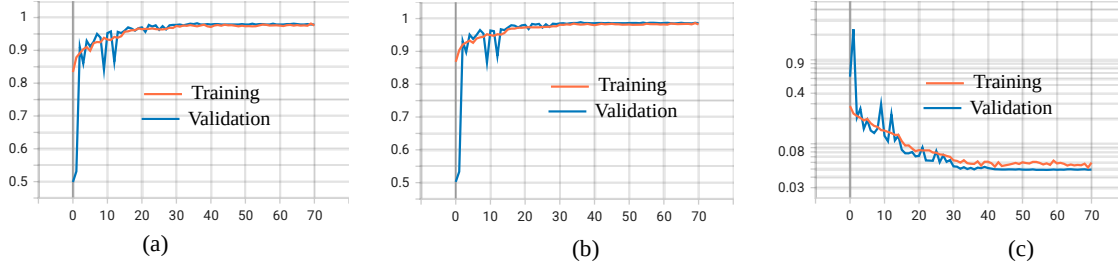


Fig. 10: Training curves of student with KD on binary classification using APTOS dataset; (a) Accuracy, (b) AUC, and (c) Loss

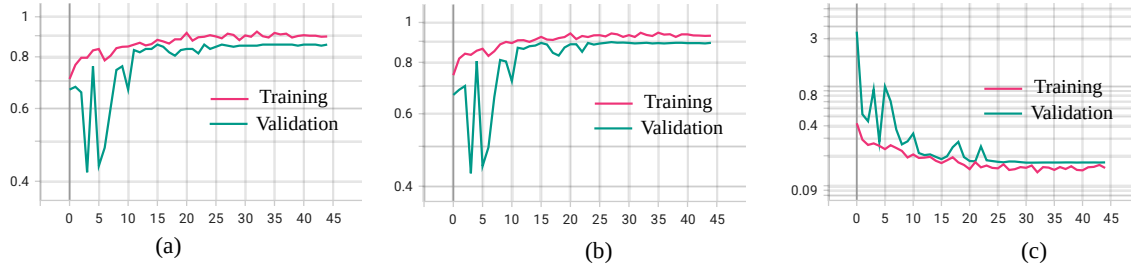


Fig. 11: Training curves of student with KD on binary classification using primary dataset; (a) Accuracy, (b) AUC, and (c) Loss

matrix in binary classification shows that our model is good at identifying the class of samples without confusion in APTOS 2019 and the primary datasets. The model predicts True Positives and True Negatives well, while False Positives and False Negatives occur more rarely. This shows that the model's predictions are very close to the actual labels. Figure 12 shows the confusion matrix of the teacher and student model with KD and without KD on each dataset.

4.3.3 Ternary Classification

The main objective of ternary classification is to build a robust model for automated grading of retinal fundus images, in which images are classified based on DR categories. This classification is crucial for the early detection and treatment of DR. The three categories considered for ternary classifications are listed as follows [70]:

1. No DR: The retina is free of detectable signs of DR, indicating a healthy retinal condition.
2. Non-proliferative DR (NPDR): This category represents the blood or liquid leakage at the back of the eye, the so-called retina. This category includes a grade of mild, moderate, and severe diabetic retinopathy.
3. Proliferative DR (PDR): This category identifies advanced DR, characterized by the growth of abnormal blood vessels. PDR poses a high risk of vision impairment if left untreated, which requires timely ophthalmologist intervention.

In our experiment, the proposed model achieves good performance across all datasets, demonstrating its robustness in ternary classification. The results are presented in Table 8 and 9 for the APTOS 2019 and Primary datasets, respectively. The performance of the model on the Primary datasets is slightly lower than on the APTOS 2019 dataset, as the Primary datasets are smaller. However, the performance of our proposed model shows comparable results to the teacher model. The proposed model achieves up to 93.09% accuracy on the APTOS 2019 dataset, whereas the proposed model achieved 85.51% accuracy on the primary dataset. The scores, particularly in recall (the ability to correctly identify all

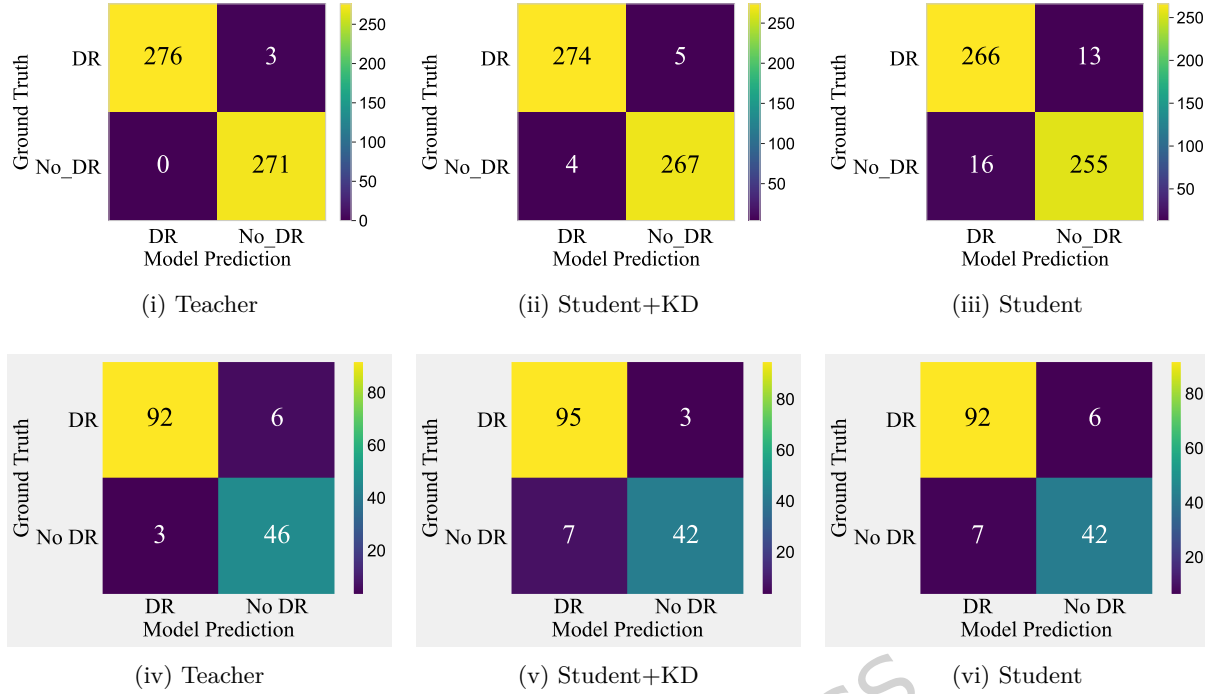


Fig. 12: Confusion matrices comparing model performance across datasets: (i-iii) APTOS 2019, (iv-vi) Primary Dataset.

relevant cases) and F1-score (which balances precision and recall), indicate reliable classification and strong generalization capability. Additionally, the model's performance is shown in detail in Figure 13. Despite a 74% reduction in trainable parameters and floating point operations per second (FLOPS), the model maintains strong performance. As our main objective is to build a model with less computational intensity while maintaining comparable performance to the *Teacher* model (pre-trained), a lightweight *Student* model is successfully developed with strong performance in ternary classification as well.

Table 8: Model's performance of ternary classification on APTOS 2019 dataset

Model	Averaging	Accuracy (%)	Precision (%)	Recall(%)	F1 Score(%)
Teacher Model	macro avg	94.18	88.58	89.34	88.93
	weighted avg	94.18	94.23	94.18	94.18
Student without KD	macro avg	79.09	67.18	69.02	66.04
	weighted avg	79.09	85.09	79.09	81.03
Student with KD (Proposed Model)	macro avg	93.09	87.37	87.32	87.34
	weighted avg	93.09	93.03	93.09	93.07

Furthermore, cross-validation is also employed for the ternary class to evaluate the class-wise performance and robustness of the proposed model as shown in Figure 10.

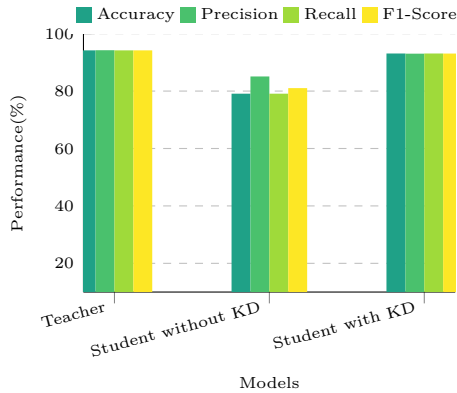
The learning curves illustrate the training patterns of our models, showing a smooth convergence for each model. Additionally, it depicts the learning behavior of the teacher model and the student models, both with and without KD. Figure 15 and Figure 16 show the learning curves of the student model with KD, whereas Figure 20 and Figure 21 shows the *Teacher* and *Student* models without KD on the APTOS 2019 and Primary datasets, respectively.

Table 9: Model's performance of ternary classification on a primary dataset

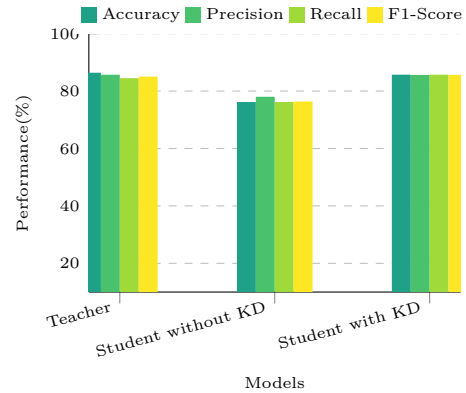
Model	Averaging	Accuracy (%)	Precision (%)	Recall(%)	F1 Score(%)
Teacher Model	macro avg	86.39	85.70	84.49	85.04
	weighted avg	86.39	86.33	86.39	86.33
Student without KD	macro avg	76.19	73.78	78.07	75.04
	weighted avg	76.19	78.01	76.19	76.36
Student with KD (Proposed Model)	macro avg	85.71	83.58	83.38	83.46
	weighted avg	85.71	85.62	85.71	85.65

Table 10: Comparison of *Student* with knowledge distillation model cross-validation on APTOS 2019, and Primary Dataset

Dataset	Fold (k)	Accuracy(%)	Precision(%)	Recall (%)	F1 Score (%)
APTOS 2019	1	87.64	89.28	87.64	88.23
	2	92.73	92.74	92.73	92.73
	3	90.55	90.26	90.55	90.38
	4	87.09	88.68	87.09	87.26
	5	93.09	93.06	93.09	93.07
	Mean	90.22	90.80	90.22	90.33
	Std	± 2.63	± 1.90	± 2.63	± 2.38
Primary Dataset	1	83.67	84.02	83.67	83.58
	2	85.03	85.77	85.03	84.63
	3	85.03	84.85	84.03	84.65
	4	85.71	85.62	85.71	85.65
	5	83.67	84.02	83.67	83.58
	Mean	84.62	84.86	84.42	84.42
	Std	± 0.88	± 0.83	± 0.89	± 0.86



(i) APTOS 2019 dataset



(ii) Primary Dataset

Fig. 13: Comparison of model performances on different datasets

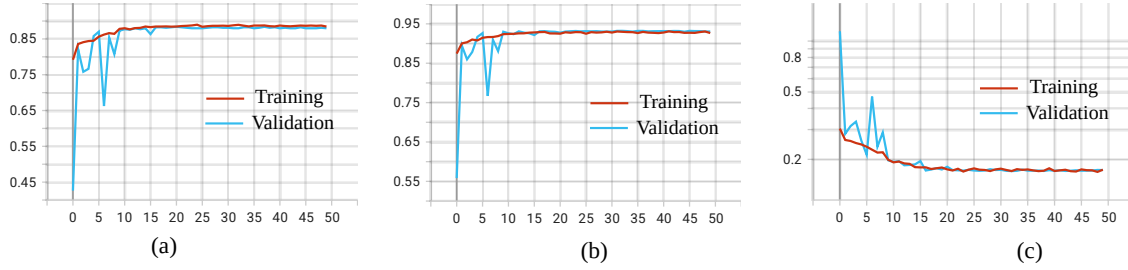


Fig. 14: Student model with KD; (a) Accuracy, (b) AUC, and (c) Loss

Fig. 15: Training curves of student with KD on ternary classification using APTOS dataset; (a) Accuracy, (b) AUC, and (c) Loss

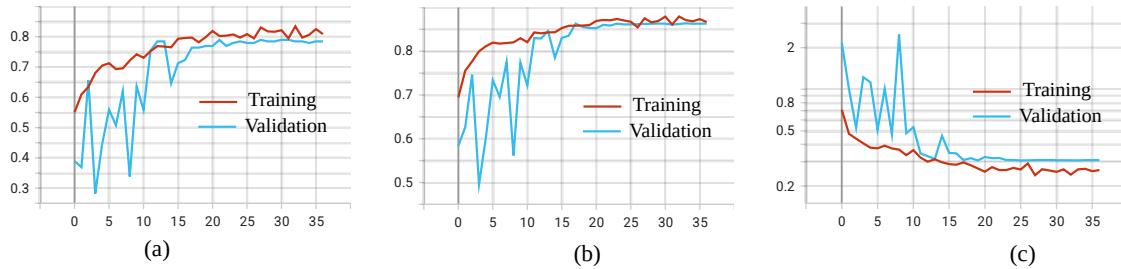


Fig. 16: Training curves of student with KD on ternary classification using primary dataset; (a) Accuracy, (b) AUC, and (c) Loss

On the other hand, the confusion matrices illustrate the classification performance of the models. The *Student* model without knowledge distillation (KD) on the APTOS 2019 dataset shows significant confusion, indicating difficulty in distinguishing between classes. The confusion matrices for each model across all datasets are presented in Figure 17. Overall, the results demonstrate that the models have effectively learned and extracted essential features and lesion patterns during training.

4.4 Performance Comparison

4.4.1 Binary Classification

The performance comparison of the proposed model with various state-of-the-art techniques is shown in Table 11. While many of the reviewed models demonstrate competitive accuracy, our proposed *Student* model with knowledge distillation (KD) achieves a superior balance between classification performance and computational efficiency.

Chetoui *et al.* [32] employed the EfficientNet-B7 architecture, achieving a recall of 98.1%. However, its large model size, approximately 66.7 million trainable parameters, renders it impractical for deployment on resource-constrained devices. In contrast, our *Student* model with KD achieves a nearly equivalent recall of 98.18% while using only 71,362 parameters, offering a drastic reduction in complexity.

Similarly, BK Anoop *et al.* [25] achieved 94.6% accuracy and 86% recall using a CNN model with an enormous 184 million parameters. Despite its high accuracy, the substantial parameter count poses limitations for scalability. Our *Student* model with KD not only surpasses this model in accuracy (98.18%) and recall (98.18%) but does so with a model that is more than 2,500 times smaller.

Bala *et al.* [28] proposed a CNN model with 1.1 million parameters, reporting 97.54% accuracy and an F1-score of 0.97. While their model is relatively lightweight compared to others, our KD-enabled *Student* model achieves comparable or better performance with less than 7% of the parameters, further emphasizing its efficiency.

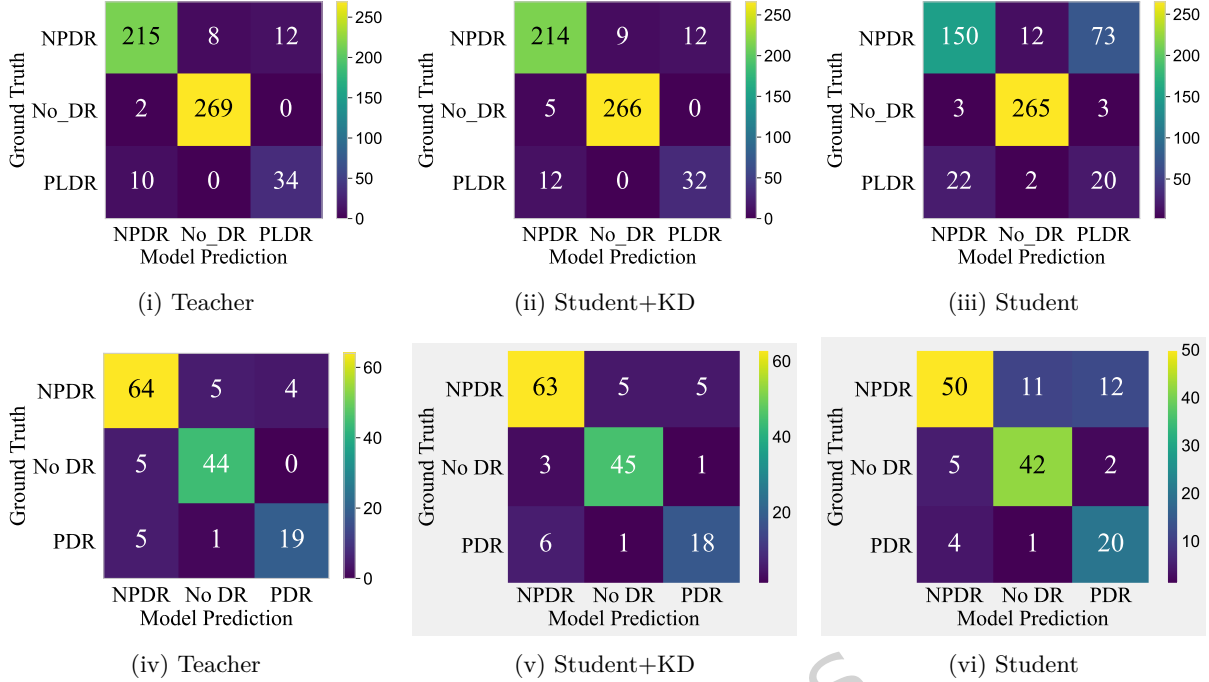


Fig. 17: Confusion matrices for ternary classification across datasets: (i-iii) APTOS 2019 dataset, (iv-vi) Primary dataset.

In another notable work, Islam *et al.* [39] utilized a hybrid ResNet152V2 + Vision Transformer (ViT) as a Teacher model, comprising 145.8 million parameters and yielding 95.15% accuracy. Our *Teacher* model, based on MobileNet, achieves a higher accuracy of 98.55% using only 279,378 parameters, underscoring our model's efficiency without compromising accuracy. Furthermore, their corresponding Student model, built using Xception with CBAM, achieved 99% accuracy with 21.4 million parameters. In comparison, our KD-based *Student* model achieves a competitive 98.18% accuracy with just 71,362 parameters.

4.4.2 Ternary Classification

The performance comparison of the proposed model with state-of-the-art techniques for ternary classification is shown in Table 12. The results demonstrate that our proposed models consistently maintain good classification performance across APTOS 2019 datasets, while significantly reducing computational requirements. Although existing studies have achieved good results, many of them involve models with a large number of trainable parameters, limiting their practicality in resource-constrained environments.

Athira *et al.* [40] utilized a ResNet50 model to achieve 94% accuracy, precision, and recall, using 25.6 million parameters. In comparison, our *Student* model with KD achieves a closely matching performance of 93% across all metrics, while requiring only 71,491 parameters, demonstrating a substantial reduction in model size.

Rao *et al.* [73] proposed an InceptionResNet model and reported 88% accuracy, precision, recall, and F1-score with 55.9 million parameters. Our *Student* model with KD outperforms this model in all metrics, with nearly 780 times fewer parameters.

Kobat *et al.* [74] used a DenseNet architecture combined with a Cubic SVM classifier, achieving 93.85% accuracy and strong precision. Butt *et al.* [41] introduced a hybrid model combining GoogleNet, ResNet-18, and SVM, reporting 89% across metrics. Although effective, these models likely carry higher computational loads compared to our student model with KD.

Our *Teacher* model, based on MobileNet, achieves 94% across all metrics on the APTOS dataset with only 279,378 parameters. Meanwhile, the *Student* model without KD shows a notable performance

Table 11: Binary classification performance on APTOS 2019 datasets

Author and Year	Models	Trainable Parameters	Accuracy	Precision	Recall	F1-Score
Chetoui <i>et al.</i> , 2020 [32]	EfficientNet-B7	66,700,000	-	-	98.1%	-
BK Anoop <i>et al.</i> , 2022 [25]	Custom CNN	184,197,154	94.6%	-	86%	-
Bala <i>et al.</i> , 2022 [28]	Custom CNN	1,100,000	97.54%	97.55%	-	0.97
Nandakumar <i>et al.</i> , 2022 [31]	Modified DenseNet-121	-	96%	93.51%	98%	0.98
Begrache <i>et al.</i> , 2023 [34]	fine-tuned Xception	-	99.8%	-	-	-
	ResNet152V2 + VIT (Teacher Model)	145,800,000	95.15%	-	-	-
Islam <i>et al.</i> , 2023 [39]	Xception + CBAM (Student Model)	21,400,000	99%	-	-	-
Tuncel <i>et al.</i> , 2025 [71]	VGG16	-	97%	97%	97%	97%
Naveen <i>et al.</i> , 2025 [72]	EffNet-SVM	-	97%	97%	97%	97%
Teacher Model	MobileNet	279,378	99.45%	99.45%	99.45%	99.45%
Student without KD	Reduced parameter MobileNet	71,362	94.73%	94.73%	94.73%	94.73%
Student with KD (Proposed Model)	Reduced parameter MobileNet	71,362	98.36%	98.36%	98.36%	98.36%

drop (70% accuracy and recall), highlighting the effectiveness of knowledge distillation in enhancing lightweight models. The KD-based *Student* model demonstrates strong performance (93%) while remaining highly efficient.

These results show that our proposed KD-based *Student* model provides a good performance and efficiency. Its lightweight architecture makes it particularly well-suited for real-time and mobile applications, where memory and processing power are often limited.

Table 12: Ternary classification performance on APTOS 2019 datasets

Author and Year	Models	Trainable Parameters	Accuracy	Precision	Recall	F1-Score
Rao <i>et al.</i> , 2020 [73]	InceptionResNet	55,900,000	88%	88%	88%	0.88
Kobat <i>et al.</i> , 2022 [74]	DenseNet + Cubic SVM	-	93.85%	90.90%	80.60%	83.78%
Butt <i>et al.</i> , 2022 [41]	GoogleNet + ResNet-18 + SVM	-	89%	89%	89%	0.89
Athira <i>et al.</i> , 2023 [40]	ResNet50	25,600,000	94%	94%	94%	0.93
Teacher Model	MobileNet	279,378	94.18%	94.23%	94.18%	94.18%
Student without KD	Reduced parameter MobileNet	71,491	79.09%	85.09%	79.09%	81.03%
Student with KD (Proposed Model)	Reduced parameter MobileNet	71,491	93.09%	93.03%	93.09%	93.07%

5 Discussion

One of the limitations of ML models is that they are heavyweight. In this regard, our proposed model is a lightweight model with comparable performance to the heavy and highly computational-intensive model proposed by researchers. Furthermore, knowledge distillation has significantly impacted the transfer of

knowledge from the teacher model to the student model. There is a huge difference between the student model built from scratch and the student model built with knowledge distillation; this shows a significant role of knowledge distillation techniques in transferring knowledge from the teacher model. Additionally, the proposed model has been evaluated on APTOS 2019, and primary datasets collected from local eye clinic centers. The performance of the model on primary datasets shows the robustness of the model on different datasets.

After a thorough experiment on the separability of NPDR severity levels, the distinction in NPDR severity levels often leads to overlaps in key image features called lesions. Furthermore, the case study presented at the TensorFlow Dev Summit 2017 [75] demonstrated that even domain experts (ophthalmologists), may provide varying gradings for the same retinal images. These grading inconsistencies are particularly challenging in cases of NPDR. Therefore, for better alignment with clinical practice, clarity, and appropriateness of medication and treatment for each category, ternary classification is chosen to be our experiment for multi-class classification.

On the other hand, t-SNE was employed to demonstrate the model's ability to cluster the test dataset effectively. We visualized the feature representations extracted from the Global Average Pooling (GAP) layer and the Output layer to assess how well the models are clustering the data. In the binary classification task, the models show a clear ability to group the test dataset according to their respective classes, demonstrated on the APTOS 2019 and primary dataset in Figure 22, 23, 24, and 25. However, in the ternary classification task, the models exhibit slight confusion when distinguishing between the three classes, particularly the *Student* model without KD. This indicates that while the model performs well in binary classification, it faces more challenges when dealing with ternary classes.

To further validate these observations, the Silhouette Score and Davies-Bouldin Index (DBI) for both binary and ternary classification on the primary dataset, and APTOS 2019 dataset are shown in Table 14, 15, 16, and 17. These metrics quantitatively assess the quality of grouping in the extracted features from GAP and the output layer of each model. The Output layer consistently achieves good feature separability compared to the GAP layer and the test dataset. Furthermore, the *Student* model with KD generally demonstrates superior clustering behavior compared to the *Student* model without KD, supporting the effectiveness of KD in improving the model's performance of feature representations. While in the binary classification, the model demonstrated better feature separability for classifying No DR and DR cases, its performance notably decreased for ternary classification, indicating the model's a little confusion in capturing lesion variations between categories like PDR and NPDR.

Knowledge distillation is a technique that enables a smaller *Student* model to learn from a larger, more complex *Teacher* model. Through this process, the *Student* model captures essential knowledge and approximates the performance of the *Teacher* model, even with reduced computational complexity. As a result, knowledge distillation produces a highly efficient *Student* model that is compact enough for deployment on resource-constrained devices. Table 13 shows the detailed model computational complexity of the teacher model and the proposed student model on the primary dataset.

Table 13: Comparison of Teacher and Student Model Computational Complexity

Metric	Teacher Model	Student Model (Proposed Model)
Computational Complexity (FLOPs)	1557.14 MFLOPS	1018.73 MFLOPS
Model Disk Size	3.92 MB	0.30 MB
RAM Usage	3.67 MB	0.21 MB

Generally, the main strengths and contributions of the study presented in this article include:

- A lightweight student model was developed from MobileNet, by the principle of simplifying the teacher network by reducing the number of network layers and the sizes of filters, as outlined by Gou *et al.* [67].
- The significance of knowledge distillation in building lightweight models with performance comparable to *Teacher* model.
- The robustness of the proposed model, demonstrated by evaluating its performance on the APTOS 2019 dataset and the primary dataset for both binary and ternary classification.

6 Conclusion and Future Work

This study presented a knowledge distillation technique to transfer knowledge from the *Teacher* model to the proposed *Student* model. The *Student* model comprises only five Depthwise Convolutional Blocks, followed by two fully connected layers for classification. With knowledge distillation, the proposed model demonstrated promising results, whereas the *Student* model without knowledge distillation struggled to perform effectively.

The models were evaluated on APTOS 2019 and primary datasets, demonstrating robustness across diverse datasets. A comparison with state-of-the-art techniques revealed that the proposed model achieves comparable performance while being significantly less resource-intensive. For binary classification, our proposed model achieved an accuracy of 98.38% on the APTOS 2019 dataset. Furthermore, the student model with knowledge distillation achieved an accuracy of 93.03% for ternary classification on APTOS 2019.

Additionally, the proposed model demonstrates strong class separability in feature space, as visualized using t-SNE on test data. To further assess separability, features extracted from the Global Average Pooling (GAP) and Output layers were visualized using t-SNE.

The findings of this study show that the proposed lightweight model with knowledge distillation achieves good performance and is suitable for deployment in resource-constrained devices. Future work will focus on enhancing the model's performance using alternative knowledge distillation techniques like FitNets, Hint-based KD, self-distillation, and using Neural Architecture Search or pruning for selecting important nodes, an algorithmic solution for inseparable classes, and employing interpretability techniques to enhance the understanding and clarity of the model's predictions.

Availability of data and material

The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Funding

This research received no funding.

Declarations

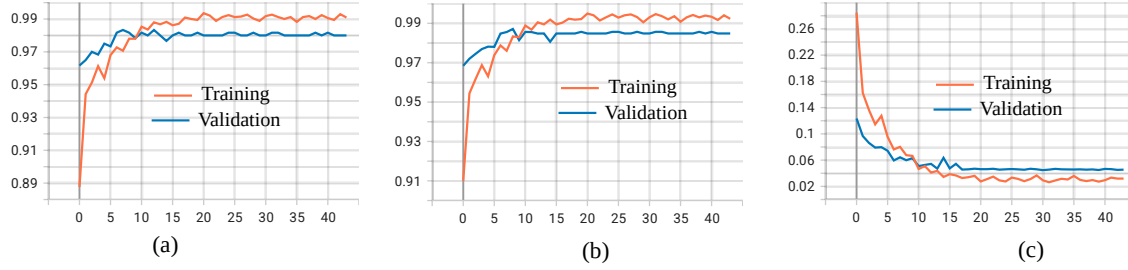
- Ethics approval: The study was conducted in accordance with the ethical guidelines established by the National Ethical Review Board (NERB). Ethical approval was granted by the Institutional Review Board of the Addis Ababa Health Bureau, with official letter reference number /A/H/6009/227, issued on 17/03/2016 (Ethiopian Calendar), corresponding to 27/11/2023 (Gregorian Calendar). Informed consent was obtained from all adult participants prior to undergoing fundoscopic imaging.
- Competing interests: The authors declare that they have no competing interests.

Author Contribution

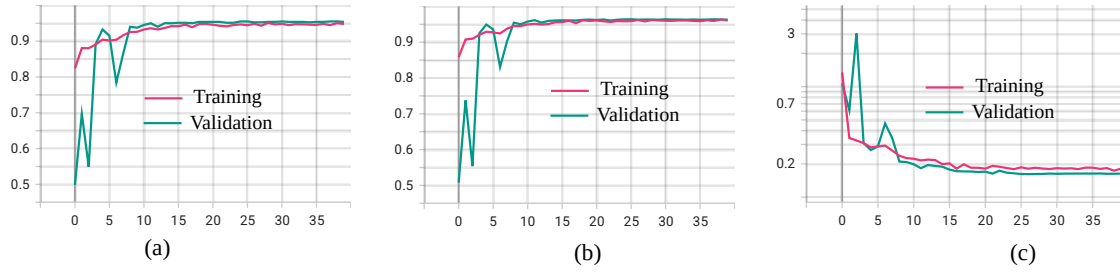
Fitsum Mesfin Dejene: Conceptualization, methodology, experimentation, and writing; Yehualashet Megersa Ayano: Conceptualization, methodology, validating experiments, and writing-editing; Degaga Wolde Feyisa: Conceptualization, methodology, and validating experiments; Taye Girma Debelee: Methodology, validating experiments, and writing-editing; Hiwot Taye Mekonnen: Data collection and supervised the data annotation process; Girum Woldegebreal Gessesse: Data collection, data annotation, trained model validation; Zelalem Chimdesa Merga: conceptualization and data collection; Hasset Tamirat Molla: Conceptualization and data collection; Destaw Mulie: Data collection, data annotation, trained model validation

Appendix A: Learning curves

See Figures 18, 19, 15, and 16.

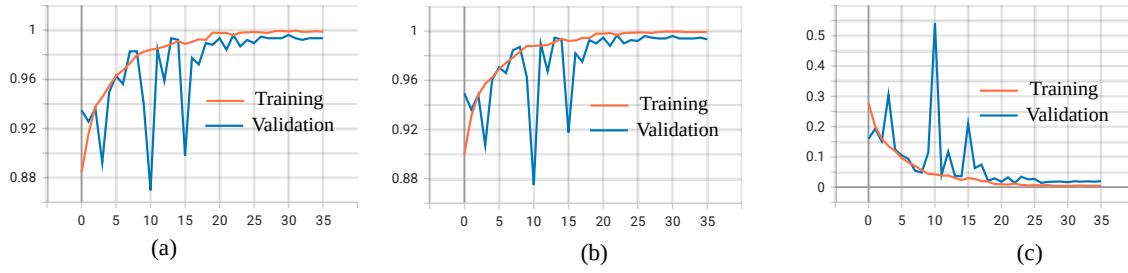


(i) Teacher model; (a) Accuracy, (b) AUC, and (c) Loss

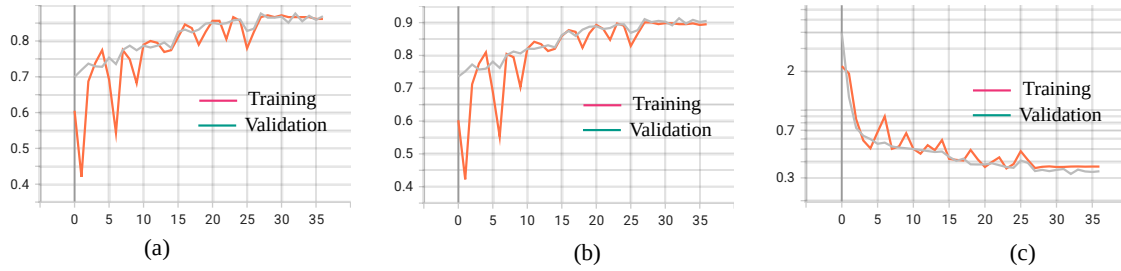


(ii) Student model without KD; (a) Accuracy, (b) AUC, and (c) Loss

Fig. 18: Training curves of binary classification on APTOS 2019 dataset

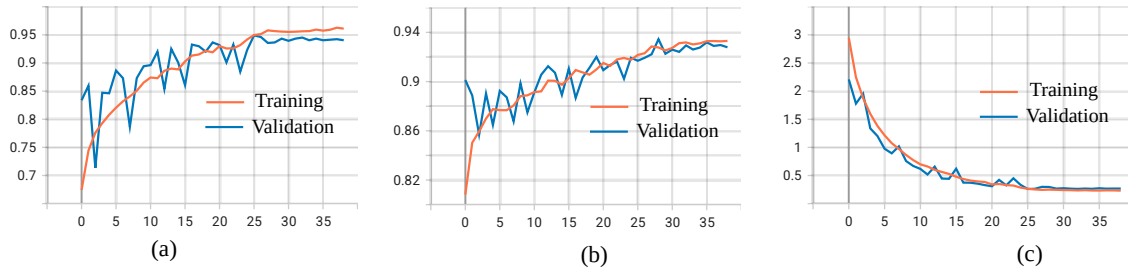


(i) Teacher model; (a) Accuracy, (b) AUC, and (c) Loss

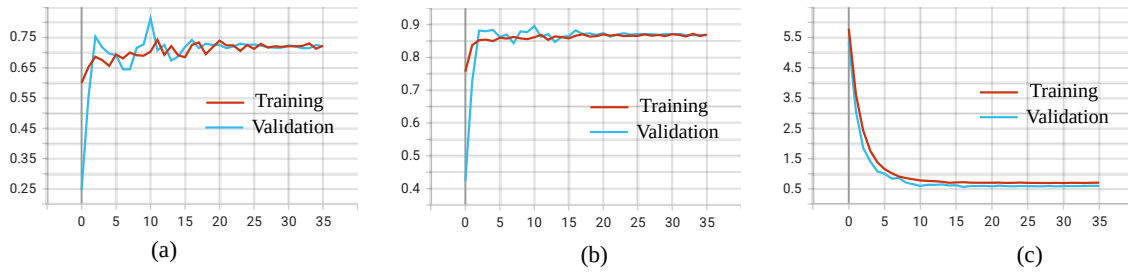


(ii) Student model without KD; (a) Accuracy, (b) AUC, and (c) Loss

Fig. 19: Training curves of binary classification on Primary Dataset

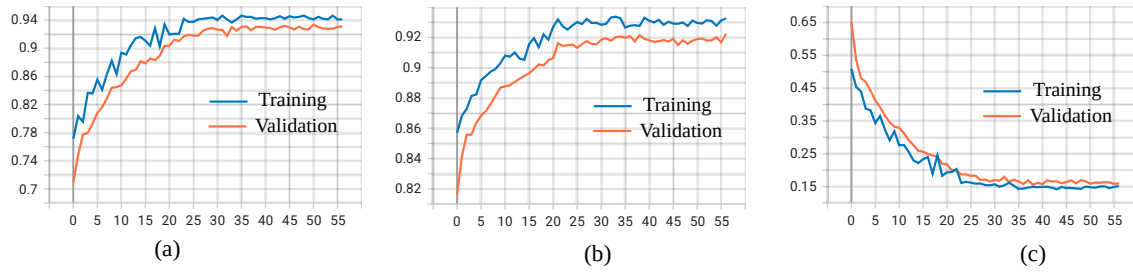


(i) Teacher model; (a) Accuracy, (b) AUC, and (c) Loss

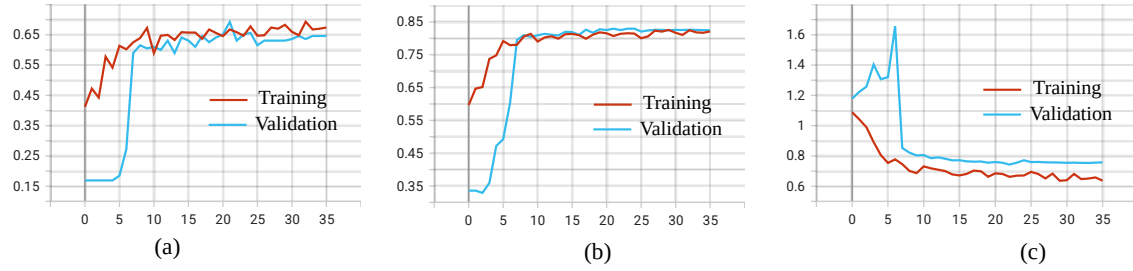


(ii) Student model without KD; (a) Accuracy, (b) AUC, and (c) Loss

Fig. 20: Ternary classification model's training curves on APTOS 2019 Dataset



(i) Teacher model; (a) Accuracy, (b) AUC, and (c) Loss



(ii) Student model without KD; (a) Accuracy, (b) AUC, and (c) Loss

Fig. 21: Ternary classification models training curves on primary dataset

Appendix B: Supplemental Figures

See Figure 22, 23, 24, 25.

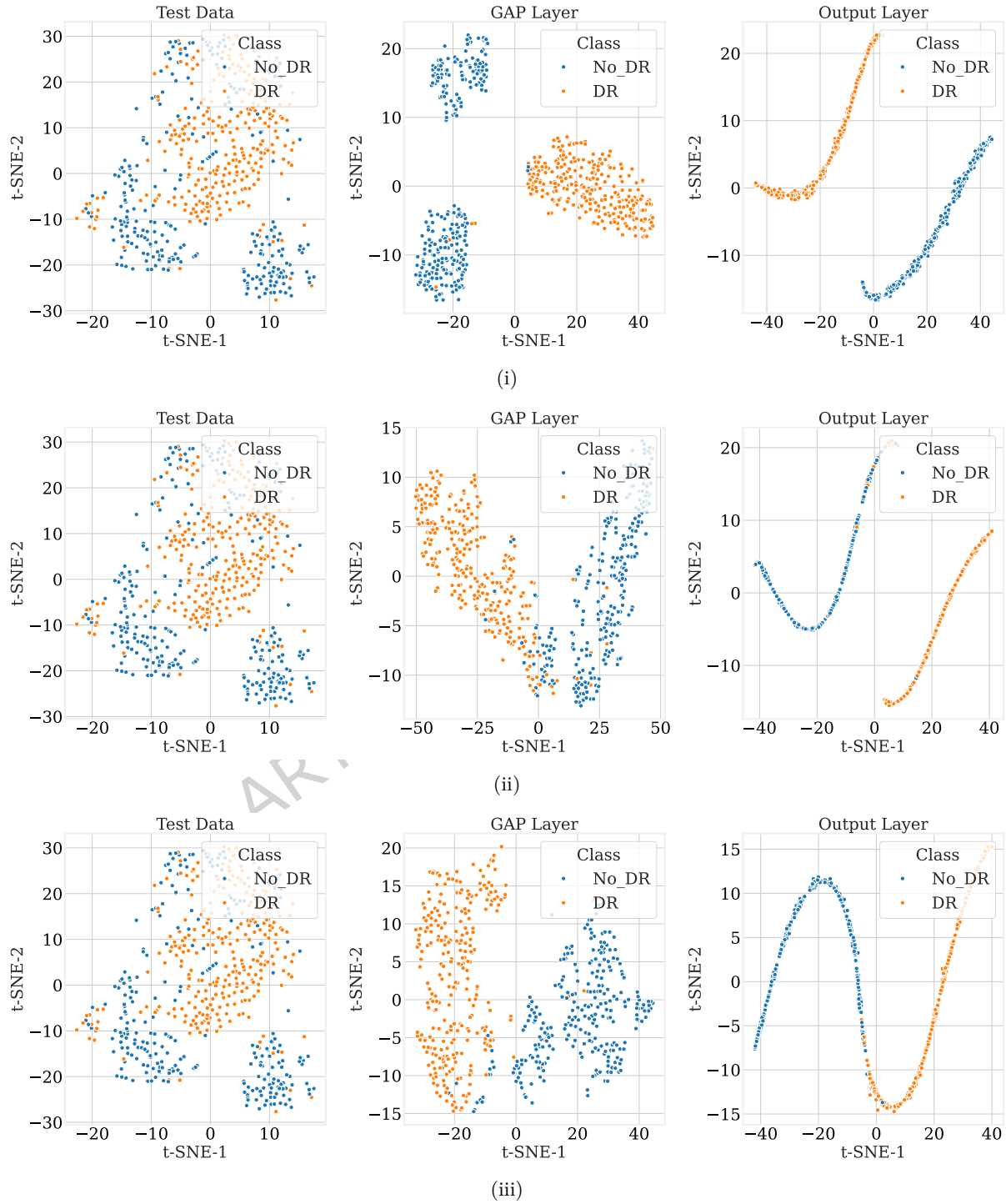


Fig. 22: t-SNE visualization of models for binary classification on APTOS 2019 dataset: (i) Teacher Model, (ii) Student Model Without KD, and (iii) Student Model With KD

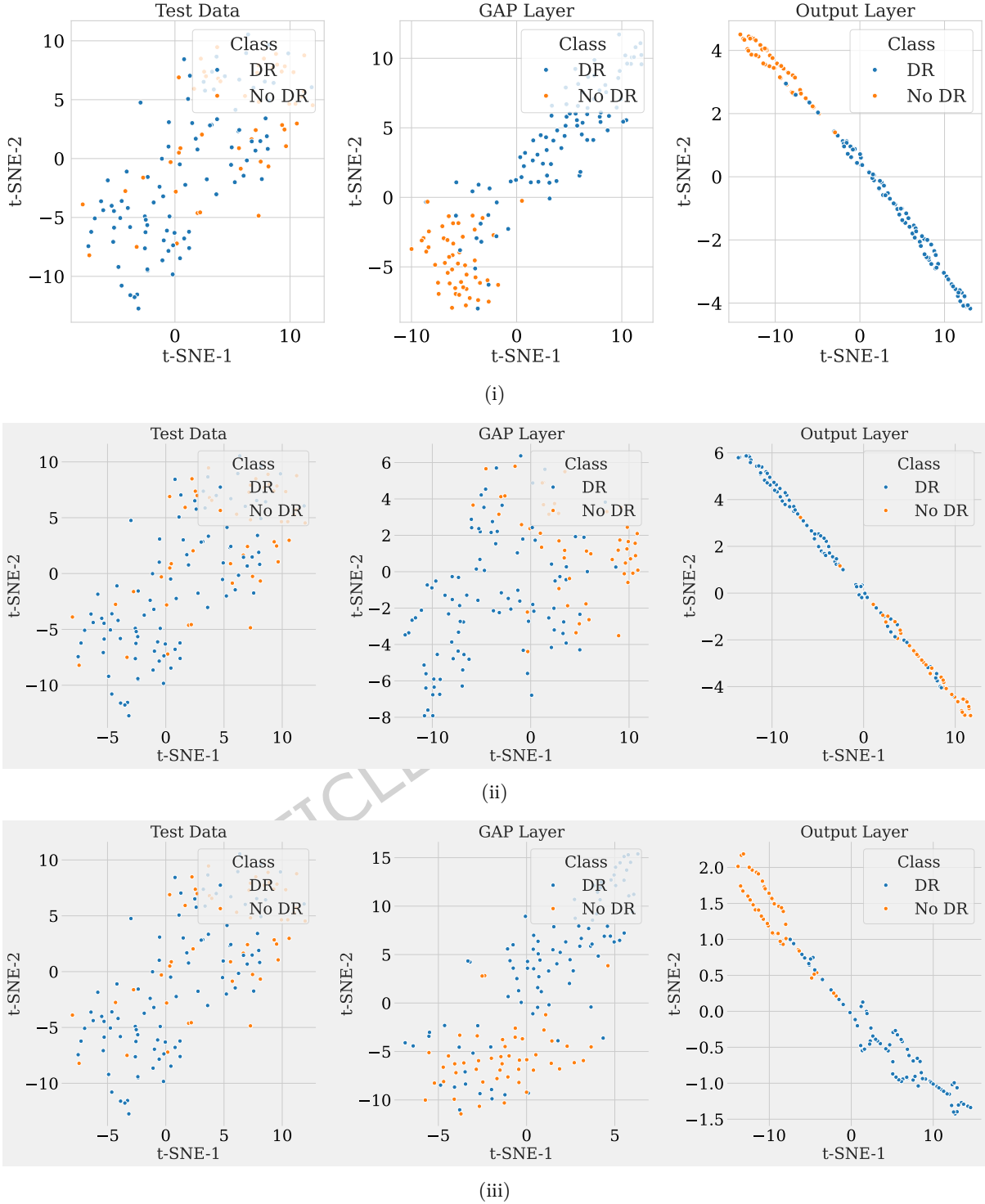


Fig. 23: t-SNE visualization of models for binary classification on primary dataset: (i) Teacher Model, (ii) Student Model Without KD, and (iii) Student Model With KD

Appendix C: Supplemental Tables

See Tables 14, 15, 16, 17

Table 14: Analysis of test data and features generated at the GAP and Output layers for clustering using Teacher and Student models (with and without knowledge distillation) for binary classification on the APTOS 2019 dataset.

Data / Layer	Model	Silhouette Score	Davies-Bouldin Index
Test Data	-	0.0373	5.71
GAP Layer	Teacher	0.3777	1.02
	Student (KD)	0.3305	1.23
	Student (w/o KD)	0.5702	0.59
Output Layer	Teacher	0.5713	0.62
	Student (KD)	0.6118	0.51
	Student (w/o KD)	0.6887	0.38

Table 15: Analysis of test data and features generated at the GAP and Output layers for clustering using Teacher and Student models (with and without knowledge distillation) for binary classification on the Primary dataset.

Data / Layer	Model	Silhouette Score	Davies-Bouldin Index
Test Data	-	0.0466	3.98
GAP Layer	Teacher	0.2254	1.14
	Student (KD)	0.1743	2.14
	Student (w/o KD)	0.1908	1.60
Output Layer	Teacher	0.5696	0.46
	Student (KD)	0.4841	0.65
	Student (w/o KD)	0.3617	0.85

Table 16: Analysis of test data and features generated at the GAP and Output layers for clustering using Teacher and Student models (with and without knowledge distillation) for ternary classification on the APTOS 2019 dataset.

Data / Layer	Model	Silhouette Score	Davies-Bouldin Index
Test Data	-	0.0567	5.53
GAP Layer	Teacher	0.1750	2.62
	Student (KD)	0.1245	3.23
	Student (w/o KD)	0.2451	5.45
Output Layer	Teacher	0.3121	1.63
	Student (KD)	0.4313	1.06
	Student (w/o KD)	0.3469	6.79

Table 17: Analysis of test data and features generated at the GAP and Output layers for clustering using Teacher and Student models (with and without knowledge distillation) for ternary classification on the Primary Dataset.

Data / Layer	Model	Silhouette Score	Davies-Bouldin Index
Test Data	-	0.0097	8.39
GAP Layer	Teacher	0.0901	2.38
	Student (KD)	0.0894	2.28
	Student (w/o KD)	0.0763	2.81
Output Layer	Teacher	0.3447	0.9433
	Student (KD)	0.2683	1.0481
	Student (w/o KD)	0.1962	1.1204

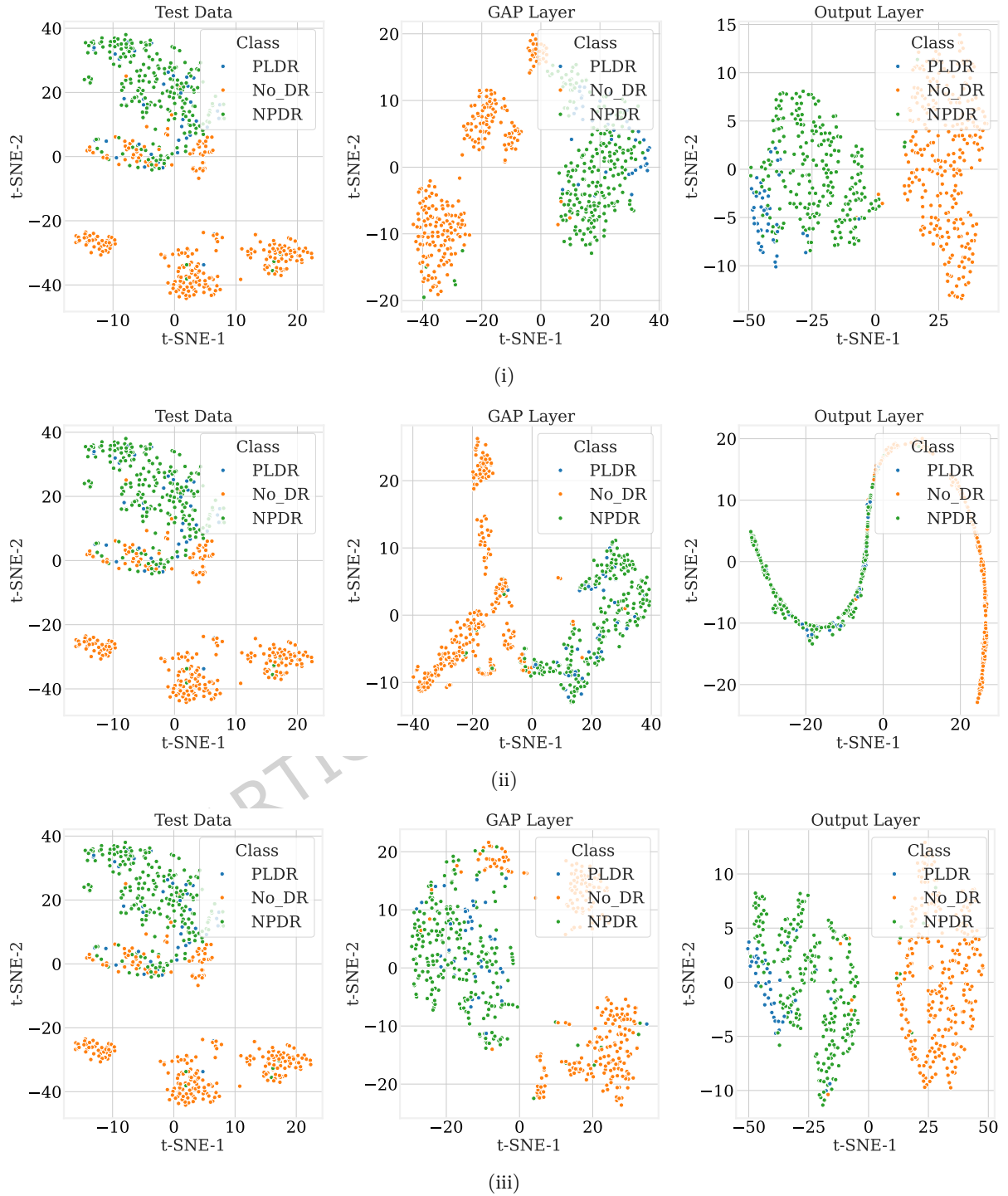


Fig. 24: t-SNE visualization of models for ternary classification on APTOS 2019 dataset: (i) Teacher Model, (ii) Student Model Without KD, and (iii) Student Model With KD

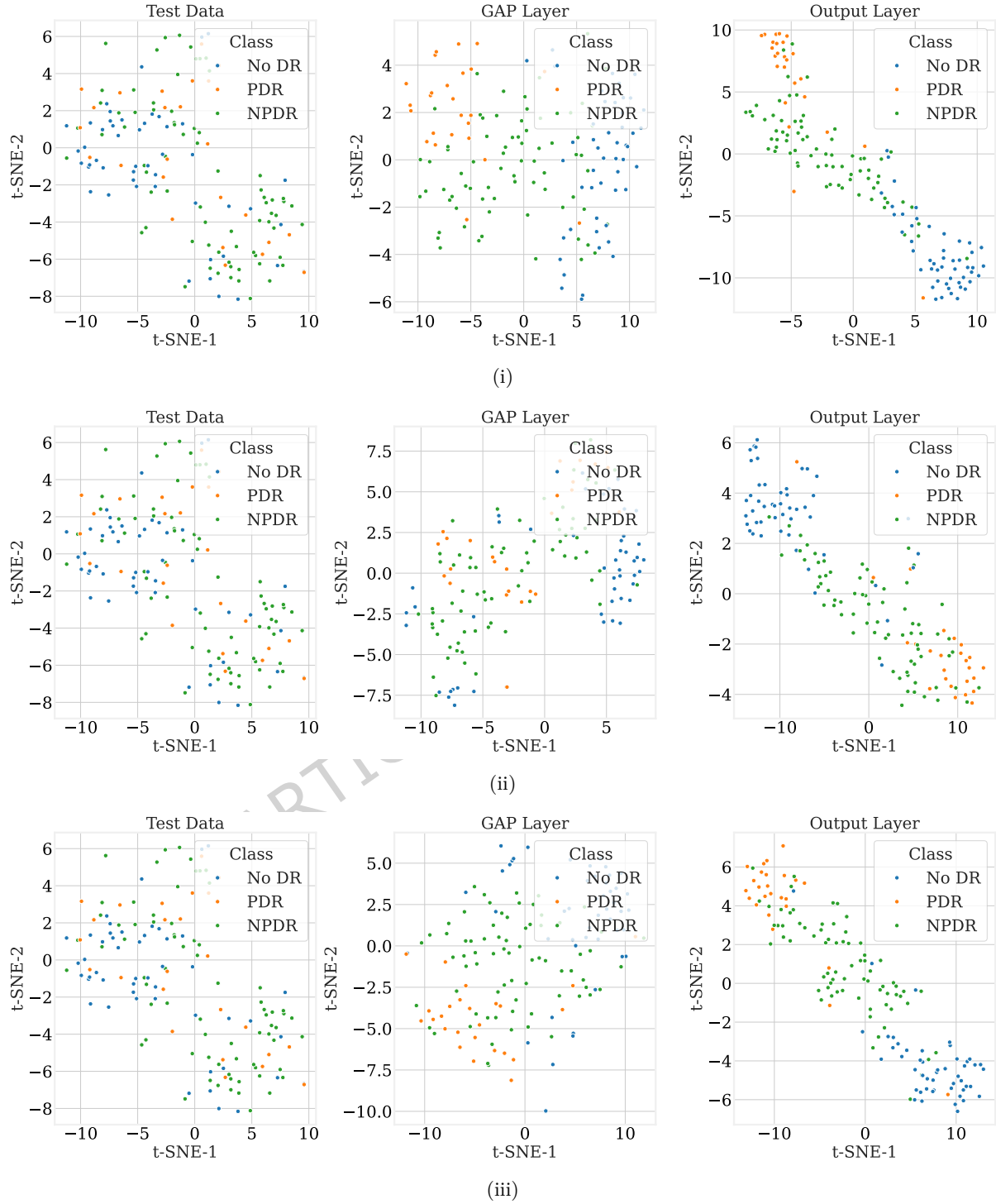


Fig. 25: t-SNE visualization of models for ternary classification on primary dataset: (i) Teacher Model, (ii) Student Model Without KD, and (iii) Student Model With KD

References

- [1] Yau, J.W., Rogers, S.L., Kawasaki, R., Lamoureux, E.L., Kowalski, J.W., Bek, T., Chen, S.-J., Dekker, J.M., Fletcher, A., Grauslund, J., *et al.*: Global prevalence and major risk factors of diabetic retinopathy. *Diabetes care* **35**(3), 556–564 (2012)
- [2] Antonetti, D.A., Klein, R., Gardner, T.W.: Mechanisms of disease diabetic retinopathy. *New England Journal of Medicine* **366**(13), 1227–1239 (2012)
- [3] Atlas, D., *et al.*: International diabetes federation. IDF Diabetes Atlas, 7th edn. Brussels, Belgium: International Diabetes Federation **33**(2) (2015)
- [4] Cho, N.H., Shaw, J., Karuranga, S., Huang, Y., Rocha Fernandes, J., Ohlrogge, A., Malanda, B.: Idf diabetes atlas: Global estimates of diabetes prevalence for 2017 and projections for 2045. *Diabetes research and clinical practice* **138**, 271–281 (2018)
- [5] Graham-Rowe, E., Lorencatto, F., Lawrenson, J., Burr, J., Grimshaw, J., Ivers, N., Presseau, J., Vale, L., Peto, T., Bunce, C., *et al.*: Barriers to and enablers of diabetic retinopathy screening attendance: a systematic review of published and grey literature. *Diabetic Medicine* **35**(10), 1308–1319 (2018)
- [6] Bishu, K.G., Jenkins, C., Yebyo, H.G., Atsbha, M., Wubayehu, T., Gebregziabher, M.: Diabetes in ethiopia: a systematic review of prevalence, risk factors, complications, and cost. *Obesity Medicine* **15**, 100132 (2019)
- [7] Fite, R.O., Lake, E.A., Hanfore, L.K.: Diabetic retinopathy in ethiopia: a systematic review and meta-analysis. *Diabetes & Metabolic Syndrome: Clinical Research & Reviews* **13**(3), 1885–1891 (2019)
- [8] Ejigu, T., Tsegaw, A.: Prevalence of diabetic retinopathy and risk factors among diabetic patients at university of gondar tertiary eye care and training center, north-west ethiopia. *Middle East African Journal of Ophthalmology* **28**(2), 71 (2021)
- [9] Kifle, M., Mbarika, V.W., Datta, P.: Telemedicine in sub-saharan africa: The case of teleophthalmology and eye care in ethiopia. *Journal of the American Society for Information Science and Technology* **57**(10), 1383–1393 (2006)
- [10] Wong, T.Y., Sun, J., Kawasaki, R., Ruamviboonsuk, P., Gupta, N., Lansingh, V.C., Maia, M., Mathenge, W., Moreker, S., Muqit, M.M., *et al.*: Guidelines on diabetic eye care: the international council of ophthalmology recommendations for screening, follow-up, referral, and treatment based on resource settings. *Ophthalmology* **125**(10), 1608–1622 (2018)
- [11] Wang, S.Y., Andrews, C.A., Gardner, T.W., Wood, M., Singer, K., Stein, J.D.: Ophthalmic screening patterns among youths with diabetes enrolled in a large us managed care network. *JAMA ophthalmology* **135**(5), 432–438 (2017)
- [12] Eppley, S.E., Mansberger, S.L., Ramanathan, S., Lowry, E.A.: Characteristics associated with adherence to annual dilated eye examinations among us patients with diagnosed diabetes. *Ophthalmology* **126**(11), 1492–1499 (2019)
- [13] Alwazae, M., Al Adel, F., Alhumud, A., Almutairi, A., Alhumidan, A., Elmorshedy, H.: Barriers for adherence to diabetic retinopathy screening among saudi adults. *Cureus* **11**(12) (2019)
- [14] Feder, R.S., Olsen, T.W., Prum, B.E., Summers, C.G., Olson, R.J., Williams, R.D., Musch, D.C.: Comprehensive adult medical eye evaluation preferred practice pattern® guidelines. *Ophthalmology* **123**(1), 209–236 (2016)

- [15] Resnikoff, S., Lansingh, V.C., Washburn, L., Felch, W., Gauthier, T.-M., Taylor, H.R., Eckert, K., Parke, D., Wiedemann, P.: Estimated number of ophthalmologists worldwide (international council of ophthalmology update): will we meet the needs? *British Journal of Ophthalmology* **104**(4), 588–592 (2020)
- [16] Dubow, M., Pinhas, A., Shah, N., Cooper, R.F., Gan, A., Gentile, R.C., Hendrix, V., Sulai, Y.N., Carroll, J., Chui, T.Y., *et al.*: Classification of human retinal microaneurysms using adaptive optics scanning light ophthalmoscope fluorescein angiography. *Investigative ophthalmology & visual science* **55**(3), 1299–1309 (2014)
- [17] Murugesan, N., Üstunkaya, T., Feener, E.P.: Thrombosis and hemorrhage in diabetic retinopathy: a perspective from an inflammatory standpoint. In: *Seminars in Thrombosis and Hemostasis*, pp. 659–664 (2015). Thieme Medical Publishers
- [18] Amin, J., Sharif, M., Yasmin, M., *et al.*: A review on recent developments for detection of diabetic retinopathy. *Scientifica* **2016** (2016)
- [19] Mateen, M., Wen, J., Nasrullah, N., Sun, S., Hayat, S.: Exudate detection for diabetic retinopathy using pretrained convolutional neural networks. *Complexity* **2020**(1), 5801870 (2020)
- [20] Senapati, A., Tripathy, H.K., Sharma, V., Gandomi, A.H.: Artificial intelligence for diabetic retinopathy detection: a systematic review. *Informatics in Medicine Unlocked*, 101445 (2024)
- [21] Shah, A., Clarida, W., Amelon, R., Hernaez-Ortega, M.C., Navea, A., Morales-Olivas, J., Dolz-Marco, R., Verbraak, F., Jorda, P.P., Heijden, A.A., *et al.*: Validation of automated screening for referable diabetic retinopathy with an autonomous diagnostic artificial intelligence system in a spanish population. *Journal of diabetes science and technology* **15**(3), 655–663 (2021)
- [22] Xie, Y., Nguyen, Q.D., Hamzah, H., Lim, G., Bellemo, V., Gunasekeran, D.V., Yip, M.Y., Lee, X.Q., Hsu, W., Lee, M.L., *et al.*: Artificial intelligence for teleophthalmology-based diabetic retinopathy screening in a national programme: an economic analysis modelling study. *The Lancet Digital Health* **2**(5), 240–249 (2020)
- [23] Poore, S., Foster, A., Zondervan, M., Blanchet, K.: Planning and developing services for diabetic retinopathy in sub-saharan africa. *International journal of health policy and management* **4**(1), 19 (2015)
- [24] Matimba, A., Woodward, R., Tambo, E., Ramsay, M., Gwanzura, L., Guramatunhu, S.: Tele-ophthalmology: opportunities for improving diabetes eye care in resource-and specialist-limited sub-saharan african countries. *Journal of telemedicine and telecare* **22**(5), 311–316 (2016)
- [25] Anoop, B., *et al.*: Binary classification of dr-diabetic retinopathy using cnn with fundus colour images. *Materials Today: Proceedings* **58**, 212–216 (2022)
- [26] Ishtiaq, U., Abdullah, E.R.M.F., Ishtiaque, Z.: A hybrid technique for diabetic retinopathy detection based on ensemble-optimized cnn and texture features. *Diagnostics* **13**(10), 1816 (2023)
- [27] Ishtiaq, U., Abdul Kareem, S., Abdullah, E.R.M.F., Mujtaba, G., Jahangir, R., Ghafoor, H.Y.: Diabetic retinopathy detection through artificial intelligent techniques: a review and open issues. *Multimedia Tools and Applications* **79**, 15209–15252 (2020)
- [28] Bala, R., Sharma, A., Goel, N.: A novel convolutional neural network architecture for diabetic retinopathy screening. In: *2022 4th International Conference on Artificial Intelligence and Speech Technology (AIST)*, pp. 1–6 (2022). IEEE

- [29] Huang, G., Liu, Z., Van Der Maaten, L., Weinberger, K.Q.: Densely connected convolutional networks. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, pp. 4700–4708 (2017)
- [30] Rêgo, S., Dutra-Medeiros, M., Soares, F., Monteiro-Soares, M.: Screening for diabetic retinopathy using an automated diagnostic system based on deep learning: diagnostic accuracy assessment. *Ophthalmologica* **244**(3), 250–257 (2021)
- [31] Nandakumar, R., Saranya, P., Ponnusamy, V., Hazra, S., Gupta, A.: Detection of diabetic retinopathy from retinal images using densenet models. *Computer Systems Science & Engineering* **45**(1) (2023)
- [32] Chetoui, M., Akhloufi, M.A.: Explainable diabetic retinopathy using efficientnet. In: 2020 42nd Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC), pp. 1966–1969 (2020). <https://doi.org/10.1109/EMBC44109.2020.9175664>
- [33] Jiang, H., Xu, J., Shi, R., Yang, K., Zhang, D., Gao, M., Ma, H., Qian, W.: A multi-label deep learning model with interpretable grad-cam for diabetic retinopathy classification. In: 2020 42nd Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC), pp. 1560–1563 (2020). <https://doi.org/10.1109/EMBC44109.2020.9175884>
- [34] Beghriche, T., Attallah, B., Brik, Y., Djerioui, M.: A multi-level fine-tuned deep learning based approach for binary classification of diabetic retinopathy. *Chemometrics and Intelligent Laboratory Systems* **237**, 104820 (2023)
- [35] Liu, P., Wang, L., Ranjan, R., He, G., Zhao, L.: A survey on active deep learning: from model driven to data driven. *ACM Computing Surveys (CSUR)* **54**(10s), 1–34 (2022)
- [36] Koch, G., Zemel, R., Salakhutdinov, R., *et al.*: Siamese neural networks for one-shot image recognition. In: ICML Deep Learning Workshop, vol. 2 (2015). Lille
- [37] Qureshi, I., Ma, J., Abbas, Q.: Diabetic retinopathy detection and stage classification in eye fundus images using active deep learning. *Multimedia Tools and Applications* **80**, 11691–11721 (2021)
- [38] Deepa, V., Sathish Kumar, C., Cherian, T.: Automated grading of diabetic retinopathy using cnn with hierarchical clustering of image patches by siamese network. *Physical and Engineering Sciences in Medicine* **45**(2), 623–635 (2022)
- [39] Islam, N., Jony, M.M.H., Hasan, E., Sutradhar, S., Rahman, A., Islam, M.M.: Toward lightweight diabetic retinopathy classification: A knowledge distillation approach for resource-constrained settings. *Applied Sciences* **13**(22), 12397 (2023)
- [40] Athira, T., Nair, J.J.: Diabetic retinopathy grading from color fundus images: An autotuned deep learning approach. *Procedia Computer Science* **218**, 1055–1066 (2023)
- [41] Butt, M.M., Iskandar, D.A., Abdelhamid, S.E., Latif, G., Alghazo, R.: Diabetic retinopathy detection from fundus images of the eye using hybrid deep learning features. *Diagnostics* **12**(7), 1607 (2022)
- [42] Barhate, N., Bhave, S., Bhise, R., Sutar, R.G., Karia, D.C.: Reducing overfitting in diabetic retinopathy detection using transfer learning. In: 2020 IEEE 5th International Conference on Computing Communication and Automation (ICCCA), pp. 298–301 (2020). IEEE
- [43] Khaled, O., El-Sahhar, M., El-Dine, M.A., Talaat, Y., Hassan, Y.M., Hamdy, A.: Cascaded architecture for classifying the preliminary stages of diabetic retinopathy. In: Proceedings of the 9th International Conference on Software and Information Engineering, pp. 108–112 (2020)

- [44] Khan, Z., Khan, F.G., Khan, A., Rehman, Z.U., Shah, S., Qummar, S., Ali, F., Pack, S.: Diabetic retinopathy detection using vgg-nin a deep learning architecture. *IEEE Access* **9**, 61408–61416 (2021)
- [45] Sudarmadji, P.W., Pakan, P.D., Dillak, R.Y.: Diabetic retinopathy stages classification using improved deep learning. In: 2020 International Conference on Informatics, Multimedia, Cyber and Information System (ICIMCIS), pp. 104–109 (2020). IEEE
- [46] Howard, A.G., Zhu, M., Chen, B., Kalenichenko, D., Wang, W., Weyand, T., Andreetto, M., Adam, H.: Mobilenets: Efficient convolutional neural networks for mobile vision applications. *arXiv preprint arXiv:1704.04861* (2017)
- [47] Riaz, H., Park, J., Choi, H., Kim, H., Kim, J.: Deep and densely connected networks for classification of diabetic retinopathy. *Diagnostics* **10**(1), 24 (2020)
- [48] Ayala, A., Ortiz Figueroa, T., Fernandes, B., Cruz, F.: Diabetic retinopathy improved detection using deep learning. *Applied Sciences* **11**(24), 11970 (2021)
- [49] Doshi, N., Oza, U., Kumar, P.: Diabetic retinopathy classification using downscaling algorithms and deep learning. In: 2020 7th International Conference on Signal Processing and Integrated Networks (SPIN), pp. 950–955 (2020). IEEE
- [50] Bhardwaj, C., Jain, S., Sood, M.: Diabetic retinopathy severity grading employing quadrant-based inception-v3 convolution neural network architecture. *International Journal of Imaging Systems and Technology* **31**(2), 592–608 (2021)
- [51] Salam, A.A., Mahadevappa, M., Das, A., Nair, M.S.: Drg-net: A graph neural network for computer-aided grading of diabetic retinopathy. *Signal, Image and Video Processing* **16**(7), 1869–1875 (2022)
- [52] Zhang, K., Pu, Z., Jin, C., Zhou, Y., Wang, Z.: A novel semi-local centrality to identify influential nodes in complex networks by integrating multidimensional factors. *Engineering Applications of Artificial Intelligence* **145**, 110177 (2025)
- [53] Bilal, A., Liu, X., Shafiq, M., Ahmed, Z., Long, H.: Nimeq-sacnet: A novel self-attention precision medicine model for vision-threatening diabetic retinopathy using image data. *Computers in biology and medicine* **171**, 108099 (2024)
- [54] Hu, T., Meng, S., Liu, C., Fang, W., Xia, Z., Hu, Y., Luo, J., Xia, X.: Lcn2 deficiency mitigates the neuroinflammatory damage following acute glaucoma. *Theranostics* **15**(7), 2967 (2025)
- [55] He, K., Chen, R., Xu, S., Ding, Y., Wu, Z., Bao, M., He, B., Li, S.: Environmental endocrine disruptor-induced mitochondrial dysfunction: a potential mechanism underlying diabetes and its complications. *Frontiers in Endocrinology* **15**, 1422752 (2024)
- [56] Sun, J., Yu, X., Li, W., Jia, B., Shi, D., Song, Y., Wang, F., Cao, B., Jiang, C.: Real-time accurate detection and analysis of breath acetone using crds: Toward metabolic dynamic monitoring and potential application. *Sensors and Actuators B: Chemical* **433**, 137422 (2025) <https://doi.org/10.1016/j.snb.2025.137422>
- [57] Yu, X., Luan, S., Lei, S., Huang, J., Liu, Z., Xue, X., Ma, T., Ding, Y., Zhu, B.: Deep learning for fast denoising filtering in ultrasound localization microscopy. *Physics in Medicine & Biology* **68**(20), 205002 (2023)
- [58] Luan, S., Yu, X., Lei, S., Ma, C., Wang, X., Xue, X., Ding, Y., Ma, T., Zhu, B.: Deep learning for fast super-resolution ultrasound microvessel imaging. *Physics in medicine & biology* **68**(24), 245023 (2023)

- [59] Liang, J., He, Y., Huang, C., Ji, F., Zhou, X., Yin, Y.: The regulation of selenoproteins in diabetes: A new way to treat diabetes. *Current Pharmaceutical Design* **30**(20), 1541–1547 (2024)
- [60] Wu, Z., Sun, W., Wang, C.: Clinical characteristics, treatment, and outcomes of pembrolizumab-induced uveitis. *Investigational New Drugs*, 1–8 (2024)
- [61] Singh, L.K., Garg, H., Pooja: Automated glaucoma type identification using machine learning or deep learning techniques. *Advancement of machine intelligence in interactive medical image analysis*, 241–263 (2020)
- [62] Qureshi, K.N., Alhudhaif, A., Qureshi, M.A., Jeon, G.: Nature-inspired solution for coronavirus disease detection and its impact on existing healthcare systems. *Computers and Electrical Engineering* **95**, 107411 (2021)
- [63] Singh, L.K., Khanna, M., Garg, H., Singh, R.: Efficient feature selection based novel clinical decision support system for glaucoma prediction from retinal fundus images. *Medical Engineering & Physics* **123**, 104077 (2024)
- [64] Singh, L.K., Garg, H., *et al.*: Detection of glaucoma in retinal images based on multiobjective approach. *International Journal of Applied Evolutionary Computation (IJAEC)* **11**(2), 15–27 (2020)
- [65] Karthik, S.D. Maggie: APTOS 2019 Blindness Detection. Kaggle. Accessed: November 27, 2025(2019). <https://kaggle.com/competitions/aptos2019-blindness-detection>
- [66] Graham, B.: Kaggle diabetic retinopathy detection competition report. University of Warwick **22**(9) (2015)
- [67] Gou, J., Yu, B., Maybank, S.J., Tao, D.: Knowledge distillation: A survey. *International Journal of Computer Vision* **129**(6), 1789–1819 (2021)
- [68] Debelee, T.G., Ayano, Y.M.: Deep learning models for audio processing applications under resource-constrained devices: A survey. In: *Pan African Conference on Artificial Intelligence*, pp. 209–232 (2022). Springer
- [69] Wang, J., Gou, L., Zhang, W., Yang, H., Shen, H.-W.: Deepvid: Deep visual interpretation and diagnosis for image classifiers via knowledge distillation. *IEEE Transactions on Visualization and Computer Graphics* **25**(6), 2168–2180 (2019) <https://doi.org/10.1109/TVCG.2019.2903943>
- [70] Memon, W.R., Lal, B., Sahto, A.A.: Diabetic retinopathy;; Frequency at level of hba1c greater than 6.5%. *The Professional Medical Journal* **24**(02), 234–238 (2017)
- [71] Tuncel, M., Uçar, M.: Classification of diabetic retinopathy disease with deep learning methods. *Artificial Intelligence Theory and Applications* **5**(1), 1–17 (2025)
- [72] Naveen, K., Anoop, B., Siju, K., Kar, M.K., Venugopal, V.: Effnet-svm: A hybrid model for diabetic retinopathy classification using retinal fundus images. *IEEE Access* (2025)
- [73] Rao, M., Zhu, M., Wang, T.: Conversion and implementation of state-of-the-art deep learning algorithms for the classification of diabetic retinopathy. *arXiv preprint arXiv:2010.11692* (2020)
- [74] Kobat, S.G., Baygin, N., Yusufoglu, E., Baygin, M., Barua, P.D., Dogan, S., Yaman, O., Celiker, U., Yildirim, H., Tan, R.-S., *et al.*: Automated diabetic retinopathy detection using horizontal and vertical patch division-based pre-trained densenet with digital fundus images. *Diagnostics* **12**(8), 1975 (2022)

- [75] Peng, L.: Case study: Tensorflow in medicine - retinal imaging. In: TensorFlow Dev Summit 2017 (2017). <https://www.youtube.com/watch?v=oOeZ7IgEN4o>

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